Advanced Sampling Methods for Household Surveys

Day 1: Foundations of Complex Sample Design

March 2–6, 2026 • Pretoria, South Africa

SADC Secretariat Member States Statistical Offices

Welcome

16 SADC Member States

United by a Common Goal:

Rigorous, Representative, Reliable Data

Your Facilitation Team

Survey Methodology

Statistical Computing

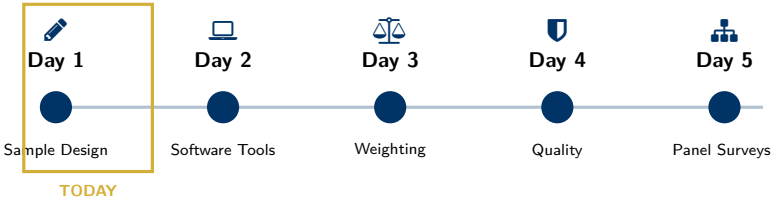
Training Design

GIS & Technology

Combining expertise from UNSD, World Bank, and National Statistics Offices

Practice + **Tools** + **Theory**

Five-Day Journey



Zambara Case

Each day follows **Lindiwe Moyo**, Chief Survey Methodologist at the Zambara National Statistics Office, as she tackles real-world sampling challenges.

Day 1 Agenda: Sample Design Foundations

- **1.1** Opening & The Challenge
- **1.2** The Design Effect
- **1.3** Multi-Stage Stratified Design
- **1.4** Optimal Allocation
- **1.5** Handling Imperfect Frames
- **1.6** Surveying Rare Populations

09:00 – 17:00 with breaks

Key Insight

Today's focus: Understanding **why** complex designs behave differently from simple random samples, and **how** to account for this in analysis.

The Challenge

A Survey Under Fire

Monday Morning at the Zambara NSO

Zambara Case

Ms. Lindiwe Moyo, newly appointed Chief Survey Methodologist at the Zambara National Statistics Office (ZNSO), arrives at work to find an urgent memo from the Minister of Planning on her desk.

"Priority: URGENT"

The Directive

Official Directive

"The 2023 Household Living Conditions Survey estimates have been challenged in Parliament. Opposition members cite NGO field data showing our **rural poverty figures may be systematically biased**.

The previous survey design has been criticized for:

- Urban over-representation
- Inadequate coverage of traditional areas
- Questionable weighting methodology

You have **seven days** to present a defensible redesign." — *Hon. Minister of Planning*

Sound familiar to your office?

What went wrong?

Urban
Bias

Frame
Decay

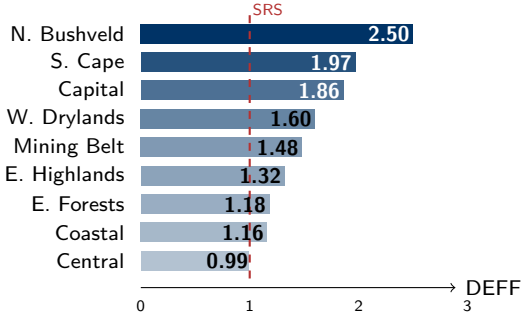
Weight
Errors

2.50

Design Effect in Northern Bushveld

Variance is **2.5 times higher** than simple random sampling would predict

The Evidence: Design Effects by Region



Key Insight

Rural regions show **highest design effects** — exactly where estimates failed.

What Does DEFF = 2.50 Mean?

Nominal Sample

50 households

DEFF = 2.5

Effective Sample

20 households

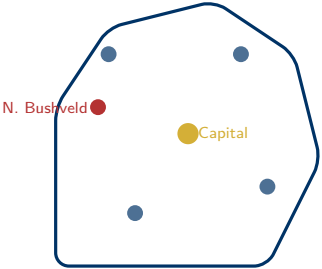
Caution

We collected 2,361 households in Northern Bushveld, but precision equals only ∼**944 independent observations**.

The Republic of Zambara

Our Case Study Nation

Zambara: A Diverse Nation



9 Regions • 248 Strata

Population: 62 million

Regions:

- 1 Metro region
- 4 Mixed regions
- 4 Rural regions

Key Challenge

Traditional areas have highest poverty but lowest survey coverage

Zambara Population Distribution

Geography Types

Zambara Case

Lindiwe discovers: Traditional areas have **30% of population** but drive **50% of poverty indicators**.

Zambara Household Survey 2024

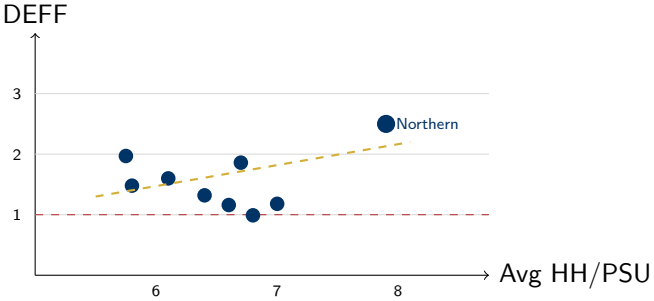
20,507	3,208	248
Households	PSU Clusters	Strata

Stratified Multi-Stage Cluster Sample with PPS

Key Variables: Household income, food security, service access, employment

Source: Zambara General Household Survey 2024 (GHS methodology)

Why Clusters Matter



Key Insight

Kish's Law: DEFF $\approx 1 + (\bar{m} - 1) \times \rho$ (Larger clusters + higher ICC = greater precision loss)

Intraclass Correlation: The Hidden Driver

Low ICC ($\rho \approx 0.02$)

Diverse incomes
Each HH adds new information

High ICC ($\rho \approx 0.20$)

Similar incomes
HHs are redundant (correlated)

Zambara Case

Northern Bushveld: Households in same village share similar economic conditions. More interviews from same PSU adds little new information.

Day 1 Learning Objectives

- Explain rationale for complex designs over simple random sampling
- Calculate and interpret design effects for stratified cluster samples
- Apply optimal allocation formulas (proportionate and Neyman)
- Develop strategies for incomplete or outdated sampling frames
- Design sampling approaches for rare and hard-to-reach populations

Each objective assessed through hands-on lab exercises

Next: Understanding Design Effects

Module 1.2

The Design Effect

Computing DEFF in R and STATA

Zambara Case

Lindiwe must now quantify the precision loss from clustering and determine which regions need the most attention in the survey redesign.

MODULE 1.2

The Design Effect

Why Clustering Costs Precision

Visual Exploration • Software Implementation

The Fundamental Question

How much precision do we lose by sampling in clusters?

Zambara Case

Lindiwe needs to quantify exactly how much the clustered design inflated her variance estimates compared to what a simple random sample would have achieved.

The **Design Effect (DEFF)** answers this question.

Design Effect: The Definition

$$\text{DEFF} = \frac{\text{Var}(\hat{\theta})_{\text{complex}}}{\text{Var}(\hat{\theta})_{\text{SRS}}}$$

Numerator

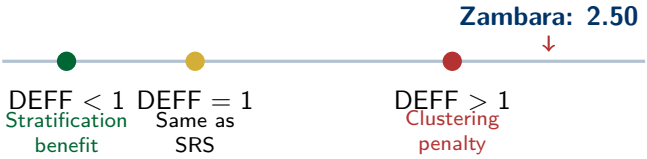
Variance under your actual complex design

Denominator

Variance if you had used simple random sampling

Source: Kish, L. (1965). *Survey Sampling*. Wiley, Chapter 8.

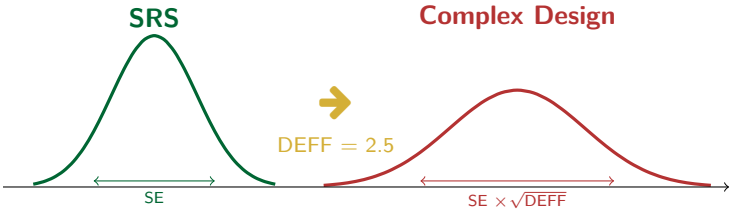
Interpreting DEFF Values



Key Insight

DEFF = 2.50 means: "The variance from my clustered sample is 2.5 times larger than what I would get from an equal-sized SRS."

Visualizing Variance Inflation



SE inflation: $\text{SE}_{\text{complex}} = \text{SE}_{\text{SRS}} \times \sqrt{\text{DEFF}} = \text{SE}_{\text{SRS}} \times 1.58$

Kish's Approximation

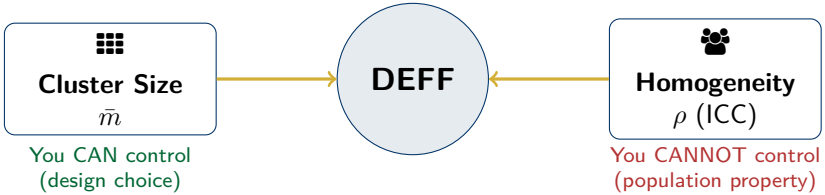
The Kish Formula

$$\text{DEFF} \approx 1 + (\bar{m} - 1) \times \rho$$

$\bar{m}$	ρ	1
Average cluster size (HH per PSU)	Intraclass correlation coefficient (ICC)	Baseline (SRS equivalent)

Source: Kish (1965); Lohr (2022), §5.5

Two Levers Control Your DEFF



Key Insight

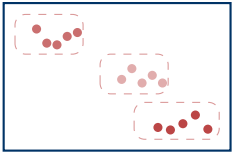
Design implication: Since you can't change ICC, reduce cluster sizes to minimize DEFF. Trade-off: More clusters = higher travel costs.

Why Does Clustering Increase Variance?

Simple Random Sample

Each unit independent
Maximum information

Cluster Sample



Units within clusters similar
Redundant information

Households in same PSU share environment → **correlated outcomes**

Intraclass Correlation Coefficient (ICC)

$$\rho = \frac{\sigma^2_{\text{between}}}{\sigma^2_{\text{between}} + \sigma^2_{\text{within}}}$$

Low ICC $(\rho \approx 0)$

Clusters are heterogeneous
Each HH adds new info
Good for precision

High ICC $(\rho \approx 0.2)$

Clusters are homogeneous
HHs within PSU redundant
Bad for precision

ICC measures within-cluster similarity relative to total variation

Deriving ICC from Observed DEFF

Inverting Kish's formula:

$$\rho = \frac{\text{DEFF} - 1}{\bar{m} - 1}$$

Zambara Case

Zambara National Level:

With DEFF $= 1.56$ and $\bar{m} = 6.4$ households per PSU:

$\rho = \frac{1.56-1}{6.4-1} = \frac{0.56}{5.4} = 0.104$

Computing DEFF

R and STATA Implementation

R: Declaring the Survey Design

Lab 1.3: R Code — Survey Design Object

```
library(survey)
library(haven)

# Load Zambara household data
zambara_hh <- read_dta("ghs-2024-hhold-v1.dta")

# Clean: remove missing income (code 9999999)
analysis_data <- zambara_hh %>%
  filter(totmhinc < 9999999, house_wgt > 0)

# Declare complex survey design
des_complex <- svydesign(
  id = ~psu,              # Cluster variable
  strata = ~stratum,      # Stratification variable
  weights = ~house_wgt,   # Survey weights
  data = analysis_data
)
```

R: Computing the Design Effect

Lab 1.3: R Code — DEFF Calculation

```
# Estimate mean income with DEFF
result <- svymean(~totmhinc, design = des_complex, deff = TRUE)

# Extract components
estimate <- coef(result)
se_complex <- SE(result)
deff_value <- deff(result)

# Display results
cat("Mean Income:  R", format(estimate, big.mark=","), "\n")
cat("SE (complex):  ", round(se_complex, 2), "\n")
cat("DEFF:          ", round(deff_value, 3), "\n")
```

Output

```
Mean Income:  R 12,617
SE (complex):   194.28
```

STATA: Declaring the Survey Design

Lab 1.3: STATA Code — svyset

```
* Load Zambara household data
use "ghs-2024-hhold-v1.dta", clear

* Clean: remove missing income values
keep if totmhinc < 9999999 & house_wgt > 0

* Declare survey design
svyset psu [pweight=house_wgt], strata(stratum) ///
    singleunit(scaled)

* Verify design
svydescribe
```

svyset stores design info; all svy: commands will use it

STATA: Computing the Design Effect

Lab 1.3: STATA Code — estat effects

```
* Estimate mean with survey design
svy: mean totmhinc

* Get design effects
estat effects
```

Output

```
Survey: Mean estimation
Number of strata =     248      Number of obs    =   20,507
Number of PSUs   =   3,208      Design df        =    2,960

--------------------------------------------------------------
             |             Linearized
             |       Mean   Std. Err.      DEFF
-------------+------------------------------------------------
    totmhinc |   12617.42     194.28     1.5634
```

Zambara: DEFF Varies by Region

Design Effects for Total Monthly Household Income

Zambara Region	PSUs	DEFF	ICC
Northern Bushveld	295	**2.50**	0.214
Southern Cape	338	1.97	0.175
Capital Region	837	1.86	0.136
Western Drylands	141	1.60	0.120
Mining Belt	234	1.48	0.096
Eastern Highlands	504	1.32	0.058
Eastern Forests	241	1.18	0.034
Coastal Plains	430	1.16	0.026
Central Plateau	188	**0.99**	<0.01

ICC derived: $\rho = (\text{DEFF} - 1)/(\bar{m} - 1)$. Source: Lab 1.3 results

R: Computing DEFF by Region

Lab 1.3: R Code — Subpopulation DEFF

```
# Loop through provinces (Zambara regions)
provinces <- 1:9

for (p in provinces) {
  # Subpopulation approach (correct variance)
  sub_result <- svymean(
    ~totmhinc,
    design = subset(des_complex, prov == p),
    deff = TRUE
  )

  deff_p <- deff(sub_result)
  cat("Province", p, ": DEFF =", round(deff_p, 2), "\n")
}
```

Use `subset()` not `filter()` to preserve design information

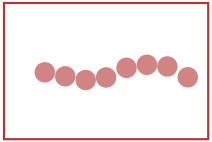
Why Rural Areas Show Higher DEFF

Urban PSU

Mixed incomes

Low ICC

Traditional PSU



Similar incomes

High ICC

Zambara Case

Northern Bushveld (DEFF = 2.50): Traditional villages share similar economic conditions — subsistence farming, limited formal employment. Each additional household interview provides little new information about village-level poverty.

Effective Sample Size: The Real Precision

$$n_{\text{eff}} = \frac{n_{\text{actual}}}{\text{DEFF}}$$

Region	**Actual n**	**DEFF**	**Effective n**
Northern Bushveld	2,361	2.50	944
Capital Region	5,413	1.86	2,910
Central Plateau	1,239	0.99	1,252
National	**20,507**	**1.56**	**13,145**

We lose **36%** of our effective sample nationally due to clustering

Module 1.2: Key Takeaways

- **DEFF** measures precision loss from complex design vs. SRS
- **Kish's formula:** DEFF $\approx 1 + (\bar{m} - 1)\rho$ links cluster size and ICC
- **ICC** is a population property; cluster size is a design choice
- Traditional/rural areas often have **higher ICC** $\rightarrow$ higher DEFF
- **Effective sample size** $=$ Actual n / DEFF — the real precision

Next: Module 1.3 — Multi-Stage Stratified Design

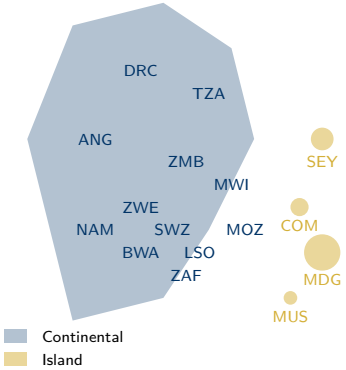
MODULE 1.3

Regional Experiences

Learning from SADC Member States

16 Nations • Shared Challenges • Collective Solutions

The SADC Region: Unity in Diversity



SADC Statistics

16 Member States

Population: 363 million
Area: 9.9 million km^2
GDP: \$721 billion

Diversity:
- 4 Island nations
- 6 Landlocked countries
- 6 Coastal mainland

Each nation brings unique sampling challenges and solutions to share.

Sampling Challenges Across SADC

Island Nations
Seychelles, Mauritius, etc.

Landlocked
Botswana, Zambia, etc.

Large/Diverse
DRC, Tanzania, Angola

Coastal Mixed
South Africa, Namibia

? What challenges does YOUR country face?

Scenario Card Exercise

Key Insight

Learning by Doing: Scenario cards describe real SADC challenges. Work in groups to propose solutions.

Format

1. Read scenario (2 min)
2. Group discussion (5 min)
3. Present solution (3 min)
4. Expert feedback (2 min)

Groups

A: Island challenges
B: Landlocked challenges
C: Large nation challenges
D: Post-conflict settings

Zambara Case

Lindiwe: "Every SADC country has solved problems others still face. Let's learn from each other."

Scenario Card #1: The Archipelago Challenge

SEYCHELLES: 115 Islands, 98,000 People

Problem: Labour Force Survey needed. 90% live on 3 islands (Mahé, Praslin, La Digue). Inter-island transport: $200-500/trip. Budget: 1,500 interviews only.

Constraints: No recent census for outer islands; some islands < 50 residents; seasonal tourism worker shifts.

? How would you design the sample?

Solution Approaches: Island Nations

Recommended Strategy

Stratified with certainty strata:

1. Main islands: Full probability sample
2. Outer islands: Two-phase (update frame via admin, then PPS)
3. Tiny islands: Cut-off sampling

Allocation

Stratum	n
Mahé (90%)	1,100
Praslin	200
La Digue	100
Outer islands	100
Total	1,500

Caution

Key: For islands, transport costs dominate. Minimize inter-island trips while maintaining coverage.

Scenario Card #2: The Distance Challenge

ZAMBIA: 752,618 km^2, 19M People

Problem: HH Budget Survey. Western Province: 5 persons/km^2. Some EAs 200+ km from roads. Rainy season (Nov-Apr) blocks access.

Constraints: 4x4 vehicles $150/day; some EAs need boats; census frame 8 years old.

? How do you handle inaccessible areas?

Solution Approaches: Remote/Sparse Areas

Adaptive Cluster Design

1. Seasonal timing (dry season)
2. Larger clusters in remote areas (8-10 HH)
3. Route optimization (group nearby PSUs)
4. Pre-identify backup PSUs

Frame Enhancement

- Satellite imagery ($0.50/km^2)
- Health facility registers
- School enrollment
- Traditional leaders

Key Insight

Zambia: Mobile data collection with offline capability for areas without network.

Scenario Card #3: The Scale Challenge

DRC: 2.3M km^2, 102M People

Problem: National HH survey needed. Country larger than Western Europe. No census since 1984. Eastern provinces in conflict. Only 2,800 km paved roads.

Constraints: 26 provinces with different conditions; active conflict zones; frame from 40-year-old census.

? Is a truly national sample even possible?

Solution Approaches: Very Large Nations

✔ Phased/Modular

1. Zone classification (stable/transitional/conflict)
2. Rolling design by quarter
3. Master sample frame
4. Accept some purposive sampling

⚠ Conflict Zones

- Document as "not covered"
- Proxy respondents (IDPs)
- Partner with humanitarian orgs
- Remote sensing estimates

Never compromise safety for coverage.

Zambara Case

DRC: "70% coverage with honest documentation beats claimed 100% with hidden biases."

Scenario Card #4: The Displacement Challenge

MOZAMBIQUE: Cabo Delgado Crisis

Problem: Poverty estimates needed. 800,000 displaced since 2017. IDP camps in neighboring provinces. 2017 census frame severely outdated.

Constraints: Security restrictions; IDPs mobile and undocumented; host communities doubled.

? How do you sample displaced populations?

Solution Approaches: Displaced Populations

Dual-Frame

Frame 1: Census (stable)

- Standard cluster sample
- Flag "new arrivals" at listing

Frame 2: Camps

- UNHCR/IOM camp lists
- Sample camps → shelters

Dynamic Updates

- Monthly camp updates
- Satellite for informal settlements
- Humanitarian coordination
- Flexible PSU boundaries

Build humanitarian relationships BEFORE survey.

Key Insight

Critical: Displaced populations are often excluded, becoming invisible in policy. Explicit inclusion essential.

Scenario Card #5: The Small Population Challenge

ESWATINI: 17,364 km^2, 1.2M People

Problem: Unemployment rates needed for 4 regions $\times$ urban/rural $\times$ sex. With 1.2M total, precision for all domains requires large sample relative to population.

Constraints: Smallest region (Lubombo) only 230K; budget max 4,000 HH; donors require RSE $<$ 10%.

? How balance national vs. domain precision?

Solution Approaches: Small Populations

Power Allocation

$$n_h = n \times \frac{\sqrt{N_h}}{\sum \sqrt{N_h}}$$

Region	Prop	Power
Hhohho	1,200	1,100
Manzini	1,400	1,150
Shiselweni	800	900
Lubombo	600	850

Small Area Est.

When direct estimates fail:

- Model-based SAE
- Borrow strength across domains
- Composite estimators

Eswatini: Direct for regions, SAE for subdomains.

Caution

With n=4,000 and 24 domains, some cells have < 100 obs. Plan for SAE from start.

Your Turn: Country Challenge Cards

Key Insight

Activity: Create a Challenge Card for your country. Exchange with another country and propose solutions.

Challenge Card Template

Country: _____

Survey type: _____

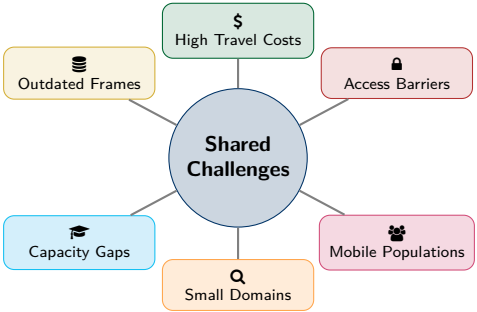
Main challenge: _____

Why difficult: _____

What you tried: _____

Time: 10 min write, 5 min exchange and discuss

Common Themes Across SADC



These challenges cut across all 16 member states. Solutions from one country can often be adapted for others.

SADC Success Stories

★ Botswana: Master Sample

Master Sample Frame across all surveys. One-time investment, 10+ year returns. 40% cost reduction.

★ South Africa: QLFS

Rotating panel: 20% rotation/quarter. Cross-sectional + longitudinal.

★ Tanzania: CAPI

Tablet-based collection with GPS. Data cleaning: 6 months → 2 weeks.

★ Namibia: Satellite

Satellite imagery for remote frame dev. Listing costs reduced 60%.

Cross-Border Survey Challenges

Zambara Case

SADC reality: Borders are porous. Populations move for work, trade, family. A Zambia survey misses Zambians in Botswana's mines.

Migration Corridors

- Zimbabwe → South Africa
- Mozambique → SA
- Malawi → Zambia
- DRC → Angola, Zambia

Harmonization

- Aligned definitions
- Coordinated reference periods
- Shared quality standards
- Joint border estimation

Key Insight

SADC Statistical Resources

References

SADC: Statistics Yearbook, RISDP, Harmonized CPI Manual

International: UN HH Survey Guidelines, WB LSMS Manual, ILO LFS Guidelines

Networks

Regional: SADC Stats Committee, AU StatCom, AFRISTAT, EASTC

Support: Stats SA, UNFPA, World Bank TFSCB

Tip: Build relationships here. Your neighbor's solution may be your answer.

Module 1.3: Key Takeaways

- SADC diversity requires flexible approaches
- Island nations: Optimize transport, accept trade-offs
- Landlocked: Seasonal timing route clustering
- Large nations: Phased, modular designs
- Peer learning: Your neighbor's solution may fit your problem

Zambara Case

Lindiwe reflects: "Zambara's challenges aren't unique. Seychelles taught us about island clustering, Zambia about seasonal timing, and DRC about honest documentation. We're stronger together."

Next: Module 1.3B — Common Sampling Pitfalls

MODULE 1.3 (Part B)

Common Sampling Pitfalls

Five Errors That Undermine SADC Surveys

Problem/Solution • Fix This Design

Five Pitfalls That Compromise Survey Quality

Pitfall 1
Ignoring Design Effects

Pitfall 2
Decayed Frames

Pitfall 3
Urban Bias

Pitfall 4
Single-Unit Strata

Pitfall 5
Non-Response Patterns

Zambara Case

Lindiwe's audit of the 2023 Zambara survey reveals all five pitfalls contributed to the parliamentary crisis. Understanding these errors is essential for designing a defensible replacement.

Interactive Learning: Fix This Design

Key Insight

For each pitfall: Identify the error → Discuss in pairs → Propose corrections → Compare with best practice

Pitfall 1: Ignoring Design Effects

The Problem

An analyst calculates confidence intervals for poverty rates using standard formulas:

$$\text{CI} = \hat{p} \pm 1.96 \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Result: "Rural poverty is 42% ± 1.8%"

Published in national report.

SRS formula used on clustered data

What's wrong here?

Pause: Discuss with your neighbor (2 minutes)

Fix This Design: Question 1

Scenario

The Zambara NSO reports: "Unemployment rate = 28.4% (SE = 0.8%)"
The analyst used `mean(unemployed)` in Excel with $n = 15{,}000$ respondents.
The actual survey used 2-stage cluster sampling with DEFF = 2.1 for employment status.

Questions for your table:

1. What is the *correct* standard error?
2. By what factor are the confidence intervals *too narrow*?
3. What policy risk does this create?

Calculate: $\text{SE}_{\text{correct}} = \text{SE}_{\text{reported}} \times \sqrt{\text{DEFF}}$

Pitfall 1: The Solution

✖ Wrong Approach

Excel / Basic Stats:

```
=AVERAGE(unemployed)
=STDEV(unemployed)/SQRT(n)
```

SE = 0.8%
95% CI: 26.8% – 30.0%

✔ Correct Approach

R with survey package:

```
svymean(~unemployed,
        design = des)
```

SE = $0.8\% \times \sqrt{2.1}$ = **1.16%**
95% CI: 26.1% – 30.7%

⚠ Caution

Consequence: CI was 45% too narrow. A 2% difference may appear significant when it isn't.

Best Practice: Always Account for Design

✔ **Rule:** Never analyze survey data without declaring the design

R: `svydesign()` | STATA: `svyset` | SPSS: Complex Samples

Checklist before analysis:

- ☑ Cluster variable identified?
- ☑ Strata variable identified?
- ☑ Weights applied?
- ☑ Design object declared?

Report requirements:

- ☑ DEFF for key estimates
- ☑ Effective sample size
- ☑ Design-corrected CIs
- ☑ Software/method stated

Pitfall 2: Decayed Sampling Frames

The Problem

The 2023 Zambara survey used the master sample from the **2013 Census**.
In 10 years:

- 847 EAs were subdivided
- 3 new townships emerged
- 12% of rural EAs "disappeared"
- Urban population grew 34%

Field teams: 18% of PSUs "not found."

2013 ⟶ 2023

10-year frame decay

Coverage error:
Who is missing?

Fix This Design: Question 2

Scenario

Your NSO must conduct a Labour Force Survey in 6 months. The only available frame is from the 2014 Census. Budget allows either:
Option A: Use 2014 frame as-is, allocate full budget to fieldwork
Option B: Spend 30% of budget on frame update activities, reduce sample by 25%

Discussion questions:

1. Which type of error does each option prioritize?
2. What frame update activities could help with limited budget?
3. When might Option A be acceptable?

Hint: Consider Total Survey Error — sampling error vs. coverage error

Pitfall 2: Frame Maintenance Strategies

Pre-Listing
Update selected PSUs before fieldwork

$ Medium

Satellite/GIS
Detect new structures via imagery analysis

$$ Higher

Admin Records
Link to utilities, voter rolls, etc.

$ Low

Key Insight

Zambara solution: Lindiwe proposes targeted pre-listing in high-growth areas (urban fringes, new settlements) combined with satellite imagery to identify PSU boundary changes. Cost: 15% of budget. Coverage improvement: estimated 8 percentage points.

Best Practice: Frame Quality Assessment

Frame Health Check — Before Every Major Survey

Indicator	Acceptable	Action Needed
Frame age	< 5 years	> 5 years
Population growth since frame	$< 10\%$	$> 10\%$
EA boundary changes	$< 5\%$	$> 5\%$
Field "not found" rate	$< 3\%$	$> 3\%$
New settlement areas	Mapped	Unknown

Zambara Case

SADC Guideline: Refresh master samples after each census. Maintain "living frame" with annual updates from admin sources.

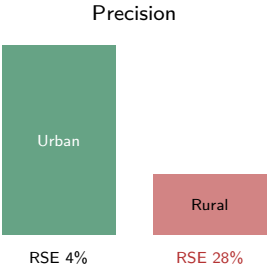
Pitfall 3: Urban Bias in Sample Allocation

⚠ The Problem

The 2023 design used **proportional allocation**:

- Urban: 60% pop → 60% sample
- Traditional: 30% pop → 30% sample
- Farms: 10% pop → 10% sample

Sounds fair, but...
Result: Traditional RSE > 25%, urban RSE < 5%.



Policy blind spot:
Can't track rural poverty

Fix This Design: Question 3

Scenario

Zambara has 9 regions. Minister requires "reliable estimates for each region."
Current proportional allocation: Capital 5,400 HH (RSE=3%), W. Drylands 280 HH (RSE=32%).
Budget: 20,000 households max.

Your task: Propose allocation achieving RSE $<$ 15% for *all* regions.

Hint: Consider: Equal allocation, Power allocation ($n_h \propto N_h^{0.5}$), or Minimum threshold + proportional

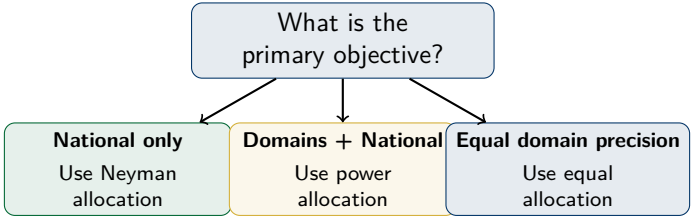
Pitfall 3: Balanced Allocation Strategies

	Proportional		Power (0.5)	
Region	n	RSE	n	RSE
Capital	5,400	3%	3,200	4%
Eastern Highlands	3,800	5%	2,800	6%
⋮	⋮	⋮	⋮	⋮
Western Drylands	280	32%	**1,100**	**14%**
Total	20,000	—	20,000	—

Key Insight

Power allocation ($n_h \propto N_h^{0.5}$) balances efficiency with domain reliability. Trade-off: National estimates slightly less precise, but all domains usable.

Best Practice: Allocation Decision Framework



Zambara Case

Lindiwe's choice: Power allocation with exponent 0.5, plus minimum floor of 800 HH/region. Ensures RSE $< 15\%$ for all regions.

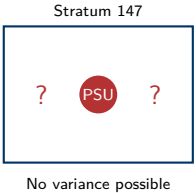
Pitfall 4: Single-Unit Strata (Lonely PSUs)

⚠ The Problem

248 strata: Province × Urban/Rural × Income.
After selection:

- 23 strata had only **1 PSU**
- Cannot estimate variance
- Software returns `NA`

"Missing SEs for 15% of estimates"



Crisis: Analysis blocked

Pitfall 4: Handling Single-Unit Strata

Four Approaches

Method	How It Works	When to Use
Collapse strata	Merge similar strata	Design stage
Certainty PSU	Self-representing	Large/unique PSUs
Centered	Assume $\bar{y}_h = \bar{Y}$	Conservative
Scaled	Average from others	STATA default

```
# R: options(survey.lonely.psu = "adjust")
* STATA: svyset psu, strata(stratum) singleunit(scaled)
```

Pitfall 5: Systematic Non-Response Patterns

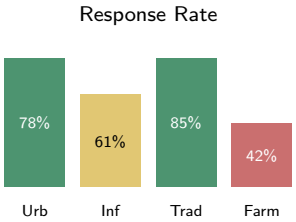
The Problem

Response rates in the 2023 survey:

- Urban formal: 78%
- Urban informal: 61%
- Traditional areas: 85%
- Farm workers: 42%

Farm workers (highest poverty) were hardest to reach — often away during work hours.

Base weights applied uniformly.



Bias direction:

Under-estimate poverty

Pitfall 5: Non-Response Adjustment Strategies

Step 1
Form adjustment cells (region × geotype)

→

Step 2
Calculate cell response rates

→

Step 3
Inflate weights: $w_{adj} = w/RR_{cell}$

R Code

```
# Response rate by cell
rr <- data %>%
  group_by(region, geotype) %>%
  summarise(rr = mean(responded))

# Adjusted weight
data$wgt_adj <- data$wgt /
  data$cell_rr
```

Farm worker example:

Original weight: 150
Cell response rate: 0.42
Adjusted weight: $150/0.42 =$ **357**

Each responding farm worker now "represents" more people.

Module 1.3: Five Pitfalls Summary

	Pitfall	Error Type	Key Fix
	Ignoring DEFF	Variance underestimate	Use `svydesign`
	Decayed frame	Coverage error	Pre-listing + GIS
	Urban bias	Domain precision	Power allocation
	Lonely PSUs	Variance undefined	Collapse/scale
	Non-response	Selection bias	Cell adjustment

Key Insight

Lindiwe's redesign addresses all five: proper design declaration, frame updates, power allocation, strata consolidation, and NR adjustment.

Next: Lab Exercise

MODULE 1.4

Multi-Stage Design

The Architecture of Complex Sample Surveys

Step-by-Step • PSU Selection • Clustering Logic

Why Multi-Stage Sampling?

✖ Simple Random Sampling

Zambara: 19M households
To sample 20,000 HH randomly:

- Need complete HH list
- Interviewers travel everywhere
- Cost: $850/interview
- Budget: $17 million

Impractical!

✔ Multi-Stage Cluster

Zambara: 3,208 PSUs
To sample 20,000 HH in clusters:

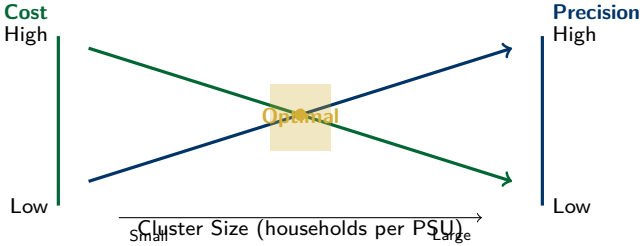
- List only selected areas
- Interviewers work locally
- Cost: $120/interview
- Budget: $2.4 million

85% cost reduction!

Key Insight

Multi-stage sampling trades statistical efficiency for operational feasibility.

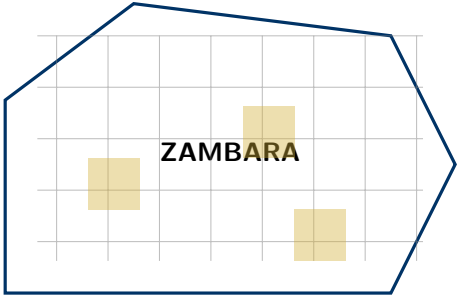
The Fundamental Trade-off



Zambara Case

Lindiwe must find the "sweet spot": clusters large enough to reduce travel costs, but small enough to maintain acceptable design effects.

Multi-Stage Design: Step 1 — Define the Frame



Frame Elements:

Enumeration Area (EA)

Primary Sampling Unit

Total EAs: 84,907
Avg HH/EA: 225
Census: 2021

Sampling Frame: Complete list of Primary Sampling Units (PSUs) with measure of size (MOS) — typically household count from most recent census.

Multi-Stage Design: Step 2 — Stratify

Stratum Level 1: Province (9 regions)

↓

Stratum Level 2: Geography Type (Urban / Traditional / Farm)

Example:
Capital Urban: 42 strata
Capital Trad: 3 strata
Bushveld Urban: 8 strata
Bushveld Trad: 15 strata
...

↓

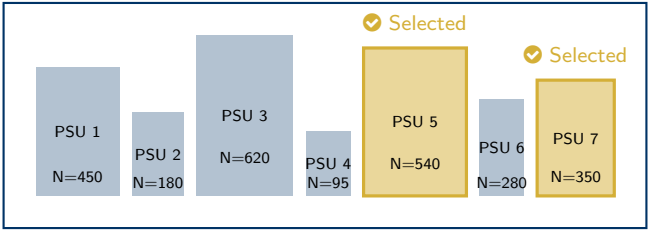
Result: 248 Strata
(Province × GeoType combinations)

Key Insight

Why stratify? Ensures representation of all subgroups, reduces variance by grouping similar units, enables domain estimation.

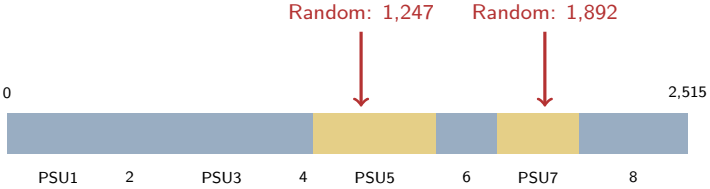
Multi-Stage Design: Step 3 — Select PSUs

Stratum: Capital Region — Urban



PSU size varies → Use **Probability Proportional to Size (PPS)**

PSU Selection: Probability Proportional to Size (PPS)



Selection probability: $\pi_i = \frac{n \times M_i}{\sum M_i}$

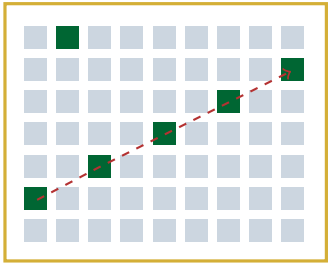
where M_i = measure of size (HH count)

Caution

Key insight: Larger PSUs have higher selection probability, but we sample *equal numbers* of households from each selected PSU → self-weighting within strata.

Multi-Stage Design: Step 4 — Select Households

Selected PSU 5 (N=540 households)



Within-PSU Selection:

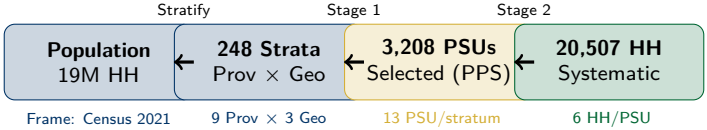
Listed HH: 540
Target sample: 6
Interval: $540/6 = 90$
Random start: 23
Select: 23, 113, 203...

Not selected

Selected (n=6)

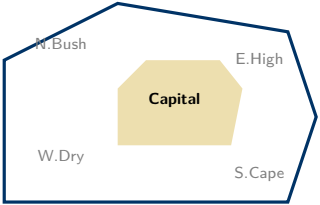
Systematic sampling within PSU: Equal probability selection using fixed interval after random start. Ensures geographic spread within cluster.

The Complete Multi-Stage Architecture



Zambara Design Parameters: $a = 3{,}208$ PSUs | $\bar{b} = 6.4$ HH/PSU | $n = 20{,}507$ HH | DEFF $= 1.56$

Zambara PSU Selection: Step 1 — Frame Setup

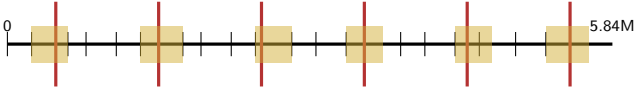


Capital Region Frame

PSU ID	EA Code	HH Count	Cumul
7001	GT-001	245	245
7002	GT-002	312	557
7003	GT-003	198	755
	...		
7842	GT-842	287	5.84M

Total PSUs in frame: 2,847
Total households: 5,844,799

Zambara PSU Selection: Step 2 — Systematic PPS

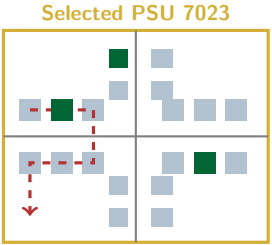


Systematic PPS Selection:
Target PSUs: $a = 837$
Interval: $I = 5{,}844{,}799/837 = 6{,}983$
Random start: $r = 2{,}451$
Select PSU containing: 2451, 9434, 16417, ...

Key Insight

Result: 837 PSUs selected from Capital Region. Each PSU's probability $\propto$ its household count.

Zambara PSU Selection: Step 3 — Household Listing



Serpentine listing path

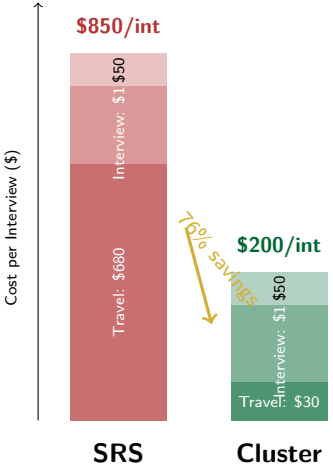
Listing Form

#	Address	Select
1	12 Main St	
2	14 Main St	
3	16 Main St	✔
4	18 Main St	
...	...	
47	8 Cross Rd	✔
...	...	

Listed: 287 HH
Sample: 6 HH
Interval: 48

Field teams list *all* dwellings in selected PSUs before systematic selection.

The Economics of Clustering



Cost Model:

$C = c_0 + a \cdot c_1 + n \cdot c_2$

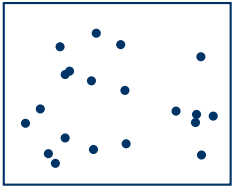
c_0 = Fixed overhead
c_1 = Cost per PSU (travel)
c_2 = Cost per interview

Zambara:

$c_1 \approx \$180$ per PSU
$c_2 \approx \$120$ per interview

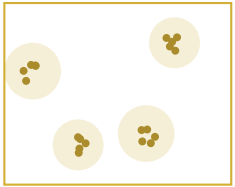
The Price of Clustering: Design Effect

Simple Random



n = 20, DEFF = 1.0
Effective n = 20

Clustered

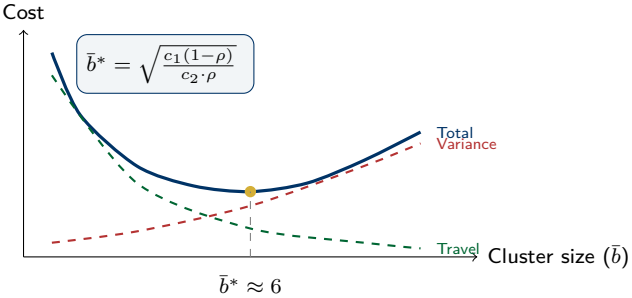


n = 20, DEFF = 1.56
Effective n = 12.8

Caution

With DEFF = 1.56, a clustered sample of 20,000 has precision of SRS of only **12,821**. Clustering penalty: 36%.

Finding the Optimal Cluster Size



Zambara Case

Zambara optimal: With $c_1/c_2 \approx 1.5$ and $\rho \approx 0.10$, optimal $\bar{b}^* \approx 6$ HH/PSU — matching actual design!

R Code: Declaring Multi-Stage Design

R: survey package

```
library(survey)

# Declare two-stage stratified cluster design
zambara_design <- svydesign(
  id = ~psu,                # Stage 1: PSU identifier
  strata = ~stratum,        # Stratification variable
  weights = ~house_wgt,     # Final survey weights
  data = zambara_hh,
  nest = TRUE               # PSUs nested within strata
)

summary(zambara_design)
```

Key: `nest=TRUE` indicates PSU IDs restart within each stratum.

STATA Code: Declaring Multi-Stage Design

STATA: svy commands

```
* Declare survey design
svyset psu [pweight=house_wgt], strata(stratum) ///
    singleunit(scaled)

* Verify design settings
svydescribe

* Now use svy: prefix for all analyses
svy: mean totmhinc
svy: proportion employed
```

Key: `singleunit(scaled)` handles lonely PSUs by scaling variance.

Zambara Design Parameters Summary

Parameter	Symbol	Zambara Value
Total population (HH)	N	19,067,953
Number of strata	H	248
PSUs in frame	—	84,907
Selected PSUs	a	3,208
Sample households	n	20,507
Mean cluster size	$\bar{b}$	6.4
Overall DEFF (income)	—	1.56
ICC (ρ)	—	0.10
Effective sample size	n_{eff}	13,145
National RSE (income)	—	1.6%
Min domain RSE	—	2.3% (Capital)
Max domain RSE		10.1% (N Bush)

Calculating Selection Probabilities

Stage 1: PSU
$$\pi_{1i} = \frac{a_h \cdot M_i}{\sum_{j \in h} M_j}$$

$\times$ →

Stage 2: HH
$$\pi_{2j|i} = \frac{b_i}{M_i}$$

$=$ →

Overall
$$\pi_{ij} = \pi_{1i} \times \pi_{2j|i}$$

Self-weighting: When b_i is constant, probability simplifies to: $\pi_{ij} = \frac{a_h \cdot b}{\sum M_j}$ — equal within stratum $\rightarrow$ weight $= 1/\pi_{ij}$

Module 1.4: Key Takeaways

- Multi-stage reduces cost 70-85
- PPS ensures proportional rep.
- Clustering increases DEFF
- Optimal $\bar{b}$ balances cost/var

Always declare design before analysis

Zambara Case

Lindiwe's progress: She understands Zambara's 2-stage stratified PPS design ($\bar{b} = 6.4$). Next: How to allocate sample across regions?

Next: Module 1.5 — Optimal Allocation Strategies

MODULE 1.4 CONTINUED

Stratification Strategy

Designing Homogeneous Groups

Province × GeoType = 248 Strata

Lindiwe's Next Challenge

Zambara Case

Lindiwe reviews the frame: "We have 84,907 EAs across Zambara. If I sample randomly, urban areas will dominate simply because they're denser. But policy makers need estimates for rural Traditional areas too. How do I ensure all population groups are represented?"

The Problem
Random sampling → Urban bias

→

The Solution
Stratified sampling

Divide population into **homogeneous groups**
Sample **independently** within each group

What is Stratification?

Definition

Stratification partitions the population into non-overlapping subgroups (strata) before sampling. Each stratum is sampled independently, ensuring representation of all groups.

Benefits

- Guaranteed coverage of all groups
- Reduced variance if strata are homogeneous
- Domain estimates with known precision
- Administrative convenience

Variance Reduction

$$V(\bar{y}_{st}) = \sum_{h=1}^{H} W_h^2 \frac{S_h^2}{n_h}$$

Stratification reduces variance by isolating **between-strata** variation.

Source: Cochran (1977), Chapter 5

Stratification Variables for Zambara

Zambara Case

Lindiwe decides: "I'll use two stratification variables that capture both geographic and settlement type variation."

Province
9 regions
(Administrative)

×

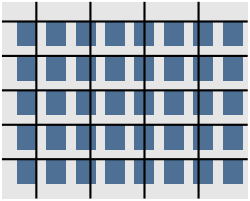
Geography Type
3 categories
(Settlement)

=

Up to 27
primary
strata

Not all combinations exist (e.g., no Farms in Capital Region) → 248 actual strata with metro subdivision

Geography Type 1: Urban



Urban Formal Settlement

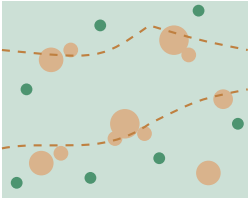
Characteristics

- Formal housing structures
- Grid street patterns
- Municipal services
- High population density
- Diverse economic activities

Zambara: 62% of population

Examples: Capital City, Mining Belt towns, Coastal Plains metros

Geography Type 2: Traditional



Traditional Settlement Area

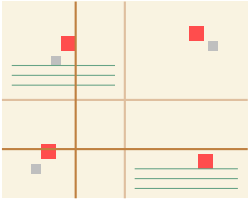
Characteristics

- Homesteads under traditional authority
- Scattered settlement pattern
- Communal land tenure
- Limited municipal services
- Agriculture-based economy

Zambara: 31% of population

Examples: Northern Bushveld villages, Eastern Highlands communal areas

Geography Type 3: Farms



Commercial Farm Area

Characteristics

- Large commercial holdings
- Very low population density
- Isolated dwelling units
- Private land ownership
- Seasonal workforce

Zambara: 7% of population

Examples: Western Drylands ranches, Central Plateau grain farms

The Stratification Matrix

Province × Geography Type (Zambara Primary Strata)

Province	Urban	Traditional	Farms
Capital Region	✓	–	–
Eastern Highlands	✓	✓	✓
Southern Cape	✓	✓	✓
Central Plateau	✓	✓	✓
Northern Bushveld	✓	✓	✓
Western Drylands	✓	–	✓
Mining Belt	✓	✓	✓
Eastern Forests	✓	✓	✓
Coastal Plains	✓	✓	✓

Not all province×geotype combinations exist. With metro subdivision: **248 final strata**.

Why Province × GeoType?

Province

Administrative relevance:

- Policy implementation at provincial level
- Budget allocation by region
- Service delivery planning
- Political accountability

Statistical: Domains of estimation required by stakeholders.

Geography Type

Socioeconomic relevance:

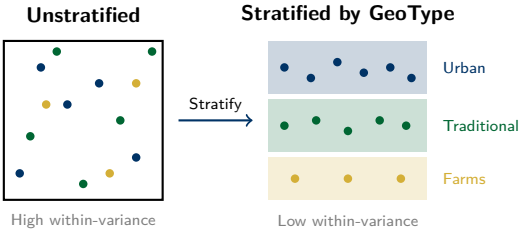
- Living conditions differ dramatically
- Service access varies
- Income sources differ
- Sampling logistics vary

Statistical: Reduces within-stratum variance significantly.

Key Insight

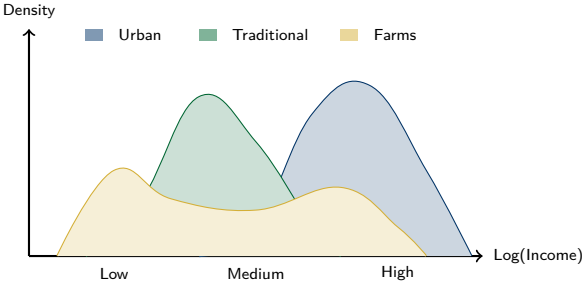
Principle: Stratify by variables that (1) define required estimation domains AND (2) correlate with key survey outcomes.

Stratum Homogeneity: Visual Evidence



Stratification moves variance **between** strata (which doesn't affect precision) rather than **within** strata.

Evidence: Income Varies by GeoType



Stratifying by GeoType captures most between-group variance in income, reducing overall sample variance.

Stratum Code Structure

3	02	0	1
Province (1-9)	Metro code	Metro (0/1)	GeoType (1-3)

Example: 30201 = Province 3 (Southern Cape), Metro 02 (Cape Metro), Non-metro (0), Urban (1)

Zambara stratum range: 10101 to 90401
Total unique strata: 248 (after removing empty combinations)

Allocation Within Strata

Zambara Case

Lindiwe asks: "Now that I have 248 strata, how do I distribute my 20,000 household sample across them?"

Proportional (Self-Weighting)

$n_h = n \times \frac{N_h}{N}$

Pros:

- Equal weights ($w = 1$ for all)
- Optimal for national estimates
- Simple implementation

Disproportional

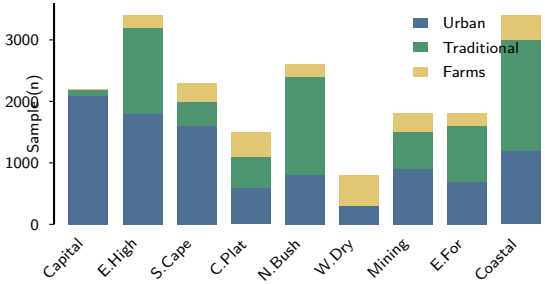
$n_h = n \times f(N_h, S_h, C_h)$

When to use:

- Domain estimates needed
- Variances differ by stratum
- Costs differ by stratum

Zambara choice: Proportional within province, slight boost for small provinces → near self-weighting.

Sample Distribution Across Strata



Sample follows population distribution with minimum constraints to ensure domain precision.

Implicit vs. Explicit Stratification

Explicit Stratification

Definition: Strata defined *before* sampling; independent samples from each.

Province × GeoType:

- Sample allocated to each stratum
- Variance estimated within strata
- Guaranteed representation

Implicit Stratification

Definition: Frame sorted by auxiliary variables; systematic sampling "spreads" sample.

Sorting by:

- EA population size
- Urban/rural within strata
- Census demographics

Key Insight

Zambara approach: Explicit stratification by Province × GeoType, then implicit stratification by sorting EAs within strata before systematic PPS selection.

R Code: Creating Strata

```r
# Create explicit strata from Province and GeoType
zambara_frame <- zambara_frame %>%
  mutate(
    # Create stratum identifier
    stratum = paste0(prov, "_", geotype),

    # Or use numeric code
    stratum_code = prov * 1000 + metro_code * 10 + geotype
  )

# Check stratum distribution
table(zambara_frame$prov, zambara_frame$geotype)

# Calculate stratum-level statistics for allocation
stratum_stats <- zambara_frame %>%
  group_by(stratum) %>%
  summarise(
    N_h = n(),                    # Frame count
    M_h = sum(hh_count),          # Total households
    S_h = sd(target_var, na.rm=T) # Stratum std dev
  )
```

Strata must be defined BEFORE sampling to enable independent selection within each.

STATA Code: Creating Strata

```
* Create explicit strata from Province and GeoType
egen stratum = group(prov geotype), label

* Alternative: numeric stratum code
gen stratum_code = prov * 1000 + metro_code * 10 + geotype

* Check stratum distribution
tab prov geotype

* Calculate stratum-level statistics
collapse (count) N_h=psu (sum) M_h=hh_count ///
         (sd) S_h=target_var, by(stratum)

* Merge back to frame for allocation
merge 1:m stratum using zambara_frame
```

In STATA, `egen group()` creates sequential stratum identifiers automatically.

Common Stratification Mistakes

✖ Mistakes to Avoid

- Too many strata → some empty or n=1
- Stratifying by outcome variable
- Ignoring frame limitations
- Creating strata *after* selection
- Single-PSU strata (variance issues)

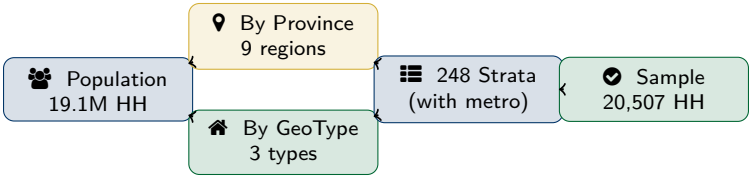
✔ Best Practices

- Minimum 2 PSUs per stratum
- Use auxiliary variables only
- Collapse small strata
- Document stratum definitions
- Test allocation before field

⚠ Caution

Rule of thumb: If a stratum has < 2 PSUs, collapse it with a similar stratum. Single-PSU strata make variance estimation impossible.

Zambara Stratification Summary



Stratification variables:	Province (9) × GeoType (3) × Metro (2)
Total strata:	248 (after removing empty combinations)
Minimum PSUs per stratum:	2 (for variance estimation)
Allocation:	Near-proportional with domain minimum constraints

Stratification: Key Takeaways

Stratification partitions popu into homogeneous groups

Province × GeoType captures key variation in Zambara

3 GeoTypes: Urban, Traditional, Farms

Reduces variance by isolating between-strata variation

Always ensure ≥2 PSUs per stratum for variance estimation

Zambara Case

Lindiwe concludes: "With 248 strata, I can guarantee representation of every Province × GeoType combination. The stratification will reduce my variance and give policy makers the domain estimates they need."

Next: Module 1.5 — Optimal Allocation Strategies

MODULE 1.4 CONTINUED

Clustering Trade-offs

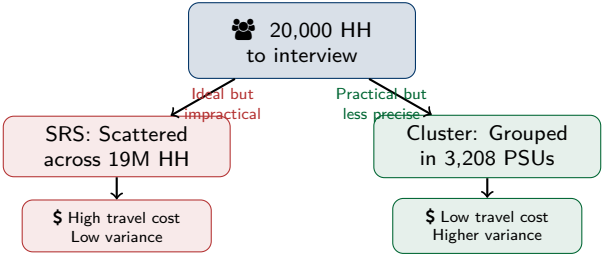
The Price of Practical Sampling

$ Cost Savings $\leftrightarrow$ Precision Loss $\rightarrow$ Optimal Balance

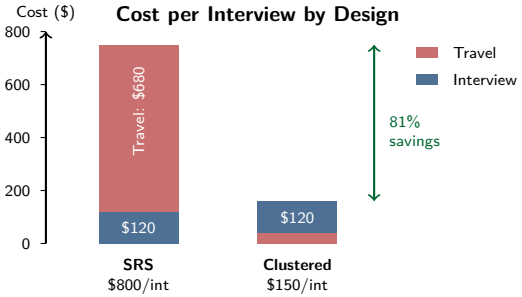
The Fundamental Dilemma

Zambara Case

Lindiwe's budget reality: "My budget allows 20,000 interviews. If I send enumerators to every corner of Zambara, travel costs will consume 80% of the budget. But if I cluster interviews geographically, I can afford more—at what cost to precision?"

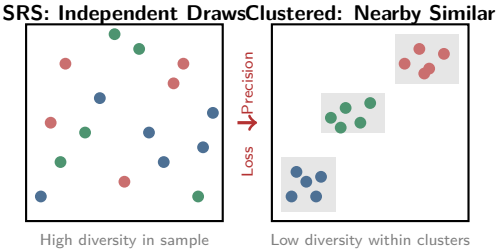


Why Clustering Saves Money



Zambara example: Clustering 6 HH per PSU reduces travel cost from $680 to $30 per interview.

But Clustering Has a Price



Key insight: Neighbors are similar. Interviewing 6 people in one village provides less information than interviewing 6 people across 6 villages.

Introducing the Design Effect (DEFF)

Design Effect

The **Design Effect** measures how much larger the variance is under the actual design compared to SRS of the same size.

$$DEFF = \frac{Var(\hat{\theta})_{complex}}{Var(\hat{\theta})_{SRS}}$$

Interpretation

- $DEFF = 1$: No effect (like SRS)
- $DEFF = 2$: Twice the variance
- $DEFF < 1$: Better than SRS (stratification gain exceeds clustering loss)

Effective Sample Size

$$n_{eff} = \frac{n}{DEFF}$$

Example: If $n = 20,000$ and $DEFF = 1.56$:

$$n_{eff} = \frac{20,000}{1.56} = 12,821$$

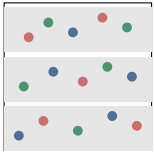
The Intraclass Correlation Coefficient (ρ)

Intraclass Correlation

The **ICC** (ρ, "rho") measures how similar units are within the same cluster compared to units in different clusters.

$$\rho = \frac{\sigma^2_{between}}{\sigma^2_{between} + \sigma^2_{within}} = \frac{\text{Variance between clusters}}{\text{Total variance}}$$

$\rho = 0$



Random mix within clusters

$\rho = 0.5$

Some similarity

$\rho = 1$

Identical within clusters

The DEFF-ICC Relationship

The Fundamental Formula:

$$\boxed{DEFF = 1 + (\bar{b} - 1) \times \rho}$$

$\bar{b}$ = **Cluster size**
Average number of units sampled per cluster (PSU)

ρ = **ICC**
Intraclass correlation (similarity within clusters)

DEFF = Design Effect
Variance inflation factor

Key Insight

Insight: DEFF increases with BOTH cluster size ($\bar{b}$) AND homogeneity (ρ). To minimize DEFF: use smaller clusters or stratify to reduce ρ.

Lab 1.3 Results: Zambara National DEFF

DEFF = 1.56

$\bar{b} = 6.4$ HH/PSU

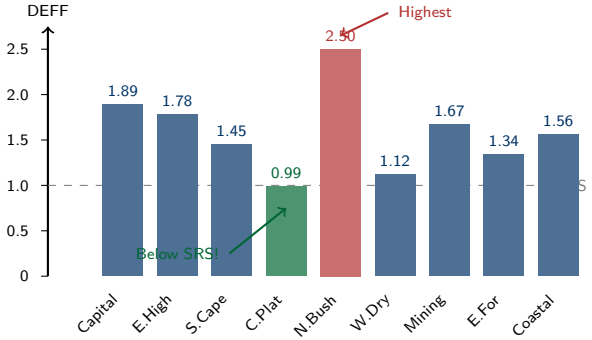
$\rho = 0.10$

Effective Sample:
$n_{eff} = \frac{20{,}507}{1.56} = 13,145$

heck: $1 + (6.4 - 1) \times 0.10 = 1.54 \approx 1.56$

Interpretation: Our 20,507 HH sample provides the precision of only 13,145 under SRS. We "lose" 36% of our sample to clustering.

Lab 1.3 Results: DEFF by Region



Note: Central Plateau (DEFF=0.99) benefits from stratification gain exceeding clustering loss. Northern Bushveld (DEFF=2.50) has high within-cluster homogeneity.

Why Does DEFF Vary by Region?

High DEFF Regions

Northern Bushveld (DEFF=2.50):

- Predominantly traditional areas
- Villages very homogeneous
- Similar income, education, housing
- High ICC ($\rho \approx 0.28$)

Low DEFF Regions

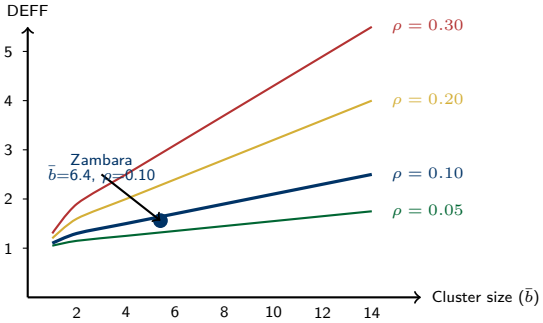
Central Plateau (DEFF=0.99):

- Mixed urban/rural
- Diverse within PSUs
- Stratification very effective
- Low ICC ($\rho \approx 0.00$)

Key Insight

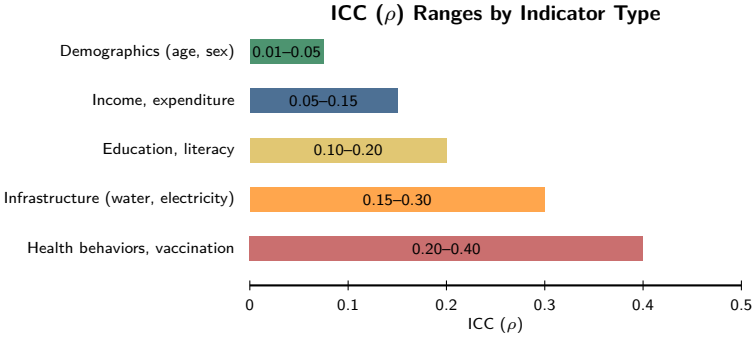
Practical implication: Different regions may warrant different cluster sizes. Consider larger clusters in diverse areas, smaller in homogeneous ones.

DEFF as Function of $\bar{b}$ and ρ



Higher ICC (ρ) means steeper DEFF increase with cluster size. Homogeneous populations suffer more from clustering.

Typical ICC Values by Variable Type



Rule of thumb: Variables influenced by geography/community tend to have higher ICC. Individual characteristics have lower ICC.

Finding the Optimal Cluster Size

Optimal Cluster Size Formula

When travel costs (c_1) and interview costs (c_2) are known: $\bar{b}^* = \sqrt{\frac{c_1(1-\rho)}{c_2 \cdot \rho}}$

Zambara Calculation

Given: c_1=\$180, c_2=\$25, ρ=0.10

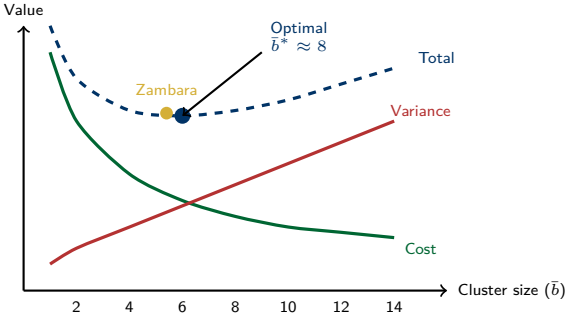
$$\bar{b}^* = \sqrt{\frac{180 \times 0.90}{25 \times 0.10}} = \sqrt{64.8} \approx 8$$

Optimal: ≈8 HH per PSU

Interpretation

Zambara uses $\bar{b}$=6.4 (slightly under optimal). Trade-off: Smaller clusters = lower DEFF but more travel. Actual choice balances multiple indicators.

The Cost-Variance Trade-off Curve



The optimal cluster size minimizes total cost (travel + precision loss). Zambara's $\bar{b} = 6.4$ is close to optimal.

Practical Implications of Clustering

What Clustering Means for Your Survey

Aspect	Without Clustering	With Clustering
Sample size (n)	20,507	20,507
Effective sample	20,507	13,145
Standard error	SE	SE $\times\sqrt{1.56} = 1.25\times$SE
Confidence interval	$\pm$1.96 SE	$\pm$2.45 SE
Travel cost	\$13.9M	\$1.8M
Cost per interview	\$800	\$150

Caution

Critical: Always account for clustering in analysis. Using SRS formulas on clustered data underestimates standard errors by 25% (for DEFF=1.56).

R Code: Calculating DEFF and ICC

```r
library(survey)

# Define complex survey design
des <- svydesign(id = ~psu, strata = ~stratum,
                 weights = ~house_wgt, data = zambara_hh)

# Calculate DEFF for income variable
svymean(~totmhinc, des, deff = TRUE)
# Output includes: mean, SE, and DEFF

# Calculate ICC from DEFF
deff <- 1.56
b_bar <- 6.4  # mean cluster size
rho <- (deff - 1) / (b_bar - 1)
# rho = 0.104

# Or use ICC package
library(ICC)
ICCest(x = psu, y = totmhinc, data = zambara_hh)
```

The `deff=TRUE` option in `svymean()` returns design effect automatically.

STATA Code: Calculating DEFF and ICC

```
* Set survey design
svyset psu [pweight=house_wgt], strata(stratum)

* Calculate mean with DEFF
svy: mean totmhinc
estat effects
* Displays DEFF for each variable

* Calculate ICC manually
loneway totmhinc psu
* Returns ICC (rho) directly

* Or compute from DEFF
local deff = 1.56
local b_bar = 6.4
local rho = (`deff' - 1) / (`b_bar' - 1)
display "ICC = " `rho'
```

STATA's `estat effects` shows DEFF, DEFT (square root), and misspecification effects.

Design Effect Decomposition

Total DEFF
1.56

Clustering Effect
$DEFF_{cluster} = 1.64$
(increases variance)

$\times$

Stratification Effect
$DEFF_{strat} = 0.95$
(decreases variance)

$1.64 \times 0.95 = 1.56$

Net effect: 56% variance inflation (clustering dominates)

Stratification partially offsets clustering loss. Without stratification, DEFF would be 1.64; with it, only 1.56.

Key Design Decisions

? What's your ρ?

- Low (<0.05)
  Cluster size matters less
- High (>0.15)
  Keep clusters small

? What's your budget?

- Tight budget
  Larger clusters, accept DEFF
- Flexible budget
  Smaller clusters, lower DEFF

Zambara: ρ=0.10 (moderate), budget-constrained $\rightarrow$ $\bar{b}$=6-8 optimal

Clustering Trade-offs: Key Takeaways

$ Clustering saves 80%+ in | But inflates variance (DEFF > 1)

DEFF $= 1 +$ | ICC (ρ) measures within-cluster similarity

Zambara: DEFF=1.56, ρ=0.10, n_{eff}=13,145 (from 20,507)

Zambara Case

Lindiwe summarizes: "Clustering is a necessary compromise. We lose 36% of our effective sample size, but save $12 million in travel costs. The key is to account for DEFF in all analyses and never use SRS formulas on clustered data."

Next: Module 1.5 — Optimal Allocation Strategies

MODULE 1.5

Allocation Strategies

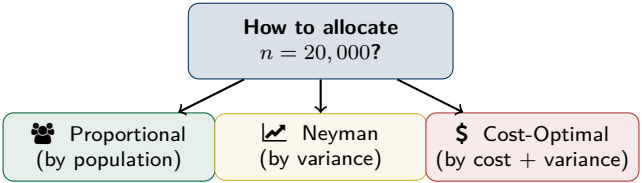
Distributing Sample Across Strata

Proportional vs. Neyman vs. $ Cost-Optimal

Lindiwe's Next Decision

Zambara Case

Lindiwe has 248 strata and 20,000 interviews. "How do I distribute samples across strata? Should large provinces get more? What about provinces with high variability? And what about the travel costs to remote areas?"



Each method optimizes for different objectives. The choice depends on survey goals and constraints.

The Allocation Problem

Sample Allocation

Given total sample size n and H strata, determine $n_1, n_2, \ldots, n_H$ such that:

$$\sum_{h=1}^{H} n_h = n \quad \text{and} \quad V(\bar{y}_{st}) = \sum_{h=1}^{H} W_h^2 \frac{S_h^2}{n_h} \text{ is minimized}$$

$W_h = N_h/N$
Stratum weight (population share)

S_h^2
Within-stratum variance of target variable

n_h
Sample size allocated to stratum h

Source: Cochran (1977), Chapter 5; Lohr (2022), Chapter 4

Method 1: Proportional Allocation

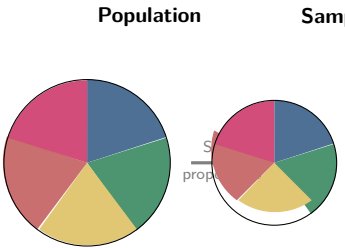
Formula

$$n_h = n \times \frac{N_h}{N} = n \times W_h$$

Sample proportional to population size

✔ Advantages

- Self-weighting design ($w_i = 1$)
- Simple to implement
- Optimal for equal stratum variances
- Easy fieldwork management



Sample mirrors population structure exactly.

Method 2: Neyman (Optimal) Allocation

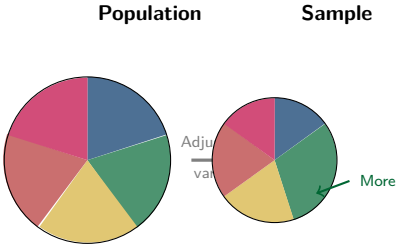
Formula

$$n_h = n \times \frac{N_h S_h}{\sum_{k=1}^{H} N_k S_k}$$

Sample proportional to $N_h \times S_h$

Key Insight

- Minimizes variance for fixed n
- Over-samples high-variance strata
- Under-samples homogeneous strata
- Requires prior variance estimates



High-variance strata get larger sample shares.

Formula Comparison: Proportional vs. Neyman

PROPORTIONAL

$$n_h = n \times \frac{N_h}{N}$$

Allocate by population size only

VS.

NEYMAN

$$n_h = n \times \frac{N_h S_h}{\sum N_k S_k}$$

Allocate by population $\times$ std dev

Key difference: Neyman multiplies by S_h — strata with higher variability get more sample

Zambara Province Characteristics

Population and Income Variability by Province

Province	**HH (000s)**	W_h **(%)**	S_h **(ZAR)**	$W_h \times S_h$	**Rel.**
Capital Region	5,120	26.8	8,450	2,265	High
Eastern Highlands	3,180	16.7	5,230	873	Med
Southern Cape	2,040	10.7	7,890	844	Med
Coastal Plains	1,890	9.9	4,120	408	Low
Northern Bushveld	1,720	9.0	3,560	320	Low
Mining Belt	1,280	6.7	6,780	454	Med
Eastern Forests	1,450	7.6	4,890	372	Low
Central Plateau	980	5.1	3,120	159	Low
Western Drylands	1,410	7.4	5,670	420	Med
Total	**19,070**	**100.0**	—	**6,115**	—

Capital Region has both the largest population AND highest income variability (S_h=8,450 ZAR).

Proportional Allocation for Zambara

Sample = 20,000 HH

Province	W_h **(%)**	n_h
Capital Region	26.8	5,360
Eastern Highlands	16.7	3,340
Southern Cape	10.7	2,140
Coastal Plains	9.9	1,980
Northern Bushveld	9.0	1,800
Mining Belt	6.7	1,340
Eastern Forests	7.6	1,520
Central Plateau	5.1	1,020
Western Drylands	7.4	1,480
Total	**100.0**	**20,000**

Calculation

Capital Region:

$$n_h = 20,000 \times \frac{5,120}{19,070}$$

$$= 20,000 \times 0.268 = 5,360$$

Result: Equal sampling fractions across all provinces ($f = n/N = 0.105\%$)

Self-weighting: All households have $w = 1$.

Neyman Allocation for Zambara

Sample = 20,000 HH

Province	$W_h S_h$	**Share**	n_h
Capital Region	2,265	37.0%	7,408
Eastern Highlands	873	14.3%	2,855
Southern Cape	844	13.8%	2,760
Coastal Plains	408	6.7%	1,334
Northern Bushveld	320	5.2%	1,046
Mining Belt	454	7.4%	1,485
Eastern Forests	372	6.1%	1,217
Central Plateau	159	2.6%	520
Western Drylands	420	6.9%	1,373
Total	**6,115**	**100%**	**20,000**

Calculation

Capital Region:

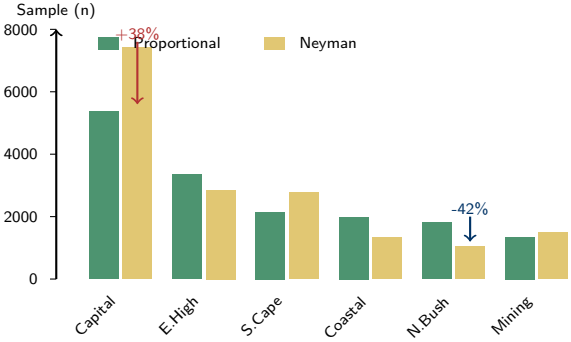
$$n_h = 20,000 \times \frac{2,265}{6,115}$$

$$= 20,000 \times 0.370 = 7,408$$

Result: Capital gets 37% of sample (vs. 27% of population)

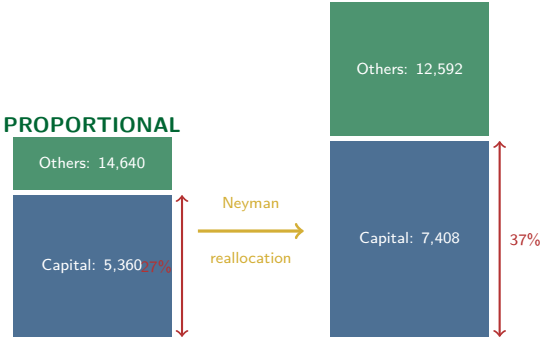
Unequal weights needed: higher-sampled strata get lower weights.

Side-by-Side: Sample Allocation Comparison



Neyman shifts sample FROM low-variance provinces (N.Bushveld) TO high-variance provinces (Capital).

Visualizing the Sample Shift

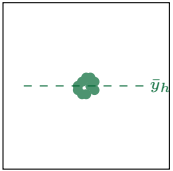


+2,048 HH to Capital
(-2,048 from others)

Neyman transfers 10% of total sample from low-variance to high-variance provinces.

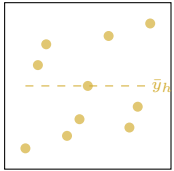
Why Over-Sample High-Variance Strata?

Low Variance (S_h small)



Few samples needed
to estimate mean precisely

High Variance (S_h large)



Many samples needed
to estimate mean precisely

Key Insight

Intuition: In homogeneous groups, a few observations tell you a lot. In heterogeneous groups, you need more observations to pin down the mean.

Variance Under Each Allocation

$$V(\bar{y}_{st}) = \sum_{h=1}^{H} W_h^2 \frac{S_h^2}{n_h}$$

PROPORTIONAL		NEYMAN
$V(\bar{y}_{st}) = 4,218$	$\longleftrightarrow$	$V(\bar{y}_{st}) = 3,741$
(in scaled units)		(in scaled units)

Neyman reduces variance by 11.3%
Equivalent to 2,260 additional interviews under proportional

The Trade-off: Weights vs. Variance

Proportional

Weights: All equal ($w_i = 1$)
Analysis: Simple, no weighting needed
Variance: Higher (not optimized)
Best for:

- Multiple indicators with different variances
- Domain estimates at stratum level
- Operational simplicity

Neyman

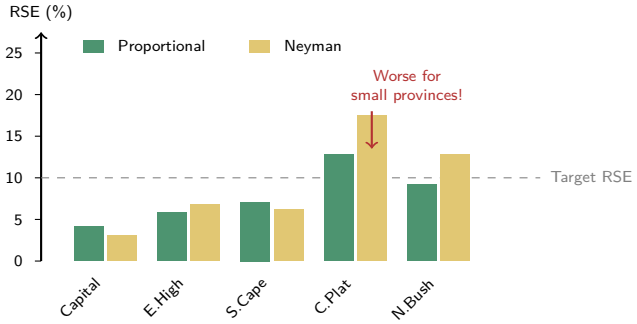
Weights: Unequal ($w_h = N_h/n_h$)
Analysis: Must use weighted estimators
Variance: Lower (optimized)
Best for:

- Single key indicator
- National-level precision priority
- Prior variance data available

Caution

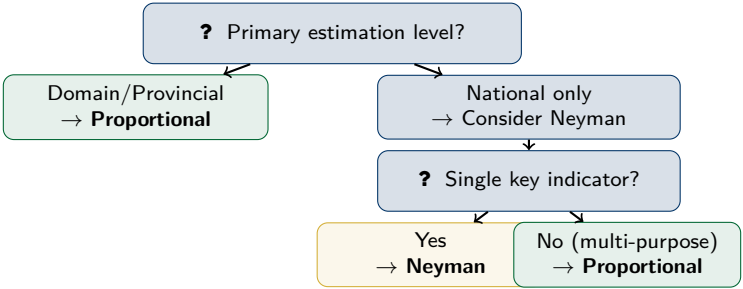
Caution: Neyman is optimal for ONE variable. For multi-purpose surveys, the "optimal" allocation for income may be suboptimal for employment.

Impact on Domain (Provincial) Estimates



Neyman improves national estimate but **harms** precision in under-sampled provinces.

Decision Guide: When to Use Each Method



Most household surveys (LFS, HES, GHS) are multi-purpose → **Proportional allocation is standard**.

R Code: Computing Allocations

```r
# Stratum data
strata <- data.frame(
  province = c("Capital", "E.Highlands", "S.Cape", ...),
  N_h = c(5120000, 3180000, 2040000, ...),  # Population
  S_h = c(8450, 5230, 7890, ...)             # Std dev of income
)

n_total <- 20000  # Total sample

# Proportional allocation
strata$W_h <- strata$N_h / sum(strata$N_h)
strata$n_prop <- round(n_total * strata$W_h)

# Neyman allocation
strata$N_S <- strata$N_h * strata$S_h
strata$n_neyman <- round(n_total * strata$N_S / sum(strata$N_S))

# Compare
strata[, c("province", "W_h", "n_prop", "n_neyman")]
```

STATA Code: Computing Allocations

```
* Input stratum data
input str20 province N_h S_h
"Capital"      5120000 8450
"E.Highlands"  3180000 5230
"S.Cape"       2040000 7890
// ... more provinces
end

local n_total = 20000

* Proportional allocation
egen N_total = total(N_h)
gen W_h = N_h / N_total
gen n_prop = round(`n_total' * W_h)

* Neyman allocation
gen N_S = N_h * S_h
egen N_S_total = total(N_S)
gen n_neyman = round(`n_total' * N_S / N_S_total)

* Compare
list province W_h n_prop n_neyman
```

Lindiwe's Allocation Decision

Zambara Case

After reviewing the options, Lindiwe decides: "The Zambara HLC Survey is multi-purpose — we measure income, employment, education, health, and housing. Different indicators have different variance patterns. Provincial estimates are required for policy. I'll use **proportional allocation with minimum constraints**."

ZAMBARA HLC SURVEY ALLOCATION

- ✔ Base: Proportional allocation (by population)
- ✔ Constraint: Minimum 800 HH per province for domain precision
- ✔ Result: Near self-weighting with small adjustments

This approach balances national precision, provincial estimates, and operational simplicity.

Allocation Strategies: Key Takeaways

⚖ **Proportional:** $n_h \propto N$

📈 **Neyman:** $n_h \propto N_h S_h$ (variance-optimal)

Neyman reduces national va

But harms domain estimates in small strata

👍 Multi-purpose surveys: Use **proportional** with minimum constraints

Coming next:
Module 1.5 Part B — Cost-optimal allocation and power allocation for small domains

Lab Exercise 1.4:
Calculate allocations for your country's provinces using the R/STATA scripts

MODULE 1.5 CONTINUED

Power Allocation

The Compromise for Multi-Level Reporting

National + Provincial = Balanced Design

The Multi-Level Reporting Challenge

Zambara Case

The Minister's requirement: "I need reliable unemployment rates for the nation AND for each province. The Western Drylands has only 7% of the population, but the Premier there demands the same precision as the Capital Region!"

National Precision (favors proportional) ⟷ Tension ⟷ **Provincial Precision (favors equal)**

? How to balance both requirements?

Two Extremes: Proportional vs. Equal Allocation

Proportional ($n_h \propto N_h$)

Best for: National estimates
Problem: Small provinces get tiny samples

Zambara example:
Central Plateau (5.1%): $n = 1,020$
$\rightarrow$ RSE $= 12.8\%$ (too high!)

Equal ($n_h = n/H$)

Best for: Domain comparisons
Problem: Wastes sample in large provinces

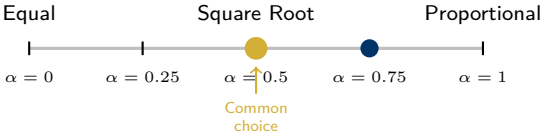
Zambara example:
All provinces: $n = 2,222$
Capital over-precise, small provinces still marginal

The Solution: Power Allocation

Power Allocation Formula

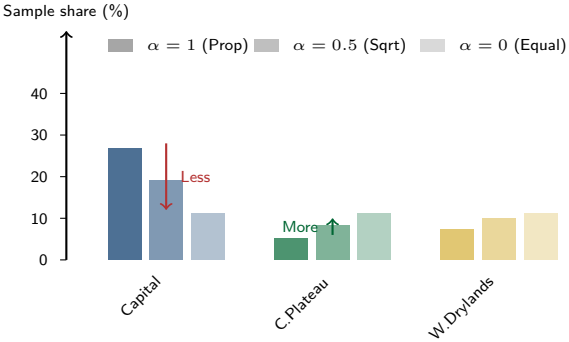
$$n_h = n \times \frac{N_h^\alpha}{\sum_{k=1}^{H} N_k^\alpha}$$

where α ("alpha") controls the allocation balance: $0 \leq \alpha \leq 1$



Key insight: $\alpha = 0.5$ (square root allocation) is a popular compromise that balances national and domain precision.

How α Shifts Sample Allocation



Lower α redistributes sample FROM large provinces TO small provinces.

Square Root Allocation ($\alpha = 0.5$)

Square Root Formula

$$n_h = n \times \frac{\sqrt{N_h}}{\sum_{k=1}^{H} \sqrt{N_k}}$$

Zambara Calculation

Capital Region:
$\sqrt{5,120,000} = 2,263$
Central Plateau:
$\sqrt{980,000} = 990$
Sum of all $\sqrt{N_h}$: 11,847

$n_{Capital} = 20,000 \times \frac{2,263}{11,847} = 3,820$

Why Square Root?

Standard error scales with $1/\sqrt{n}$
To equalize *relative* precision across domains, allocate proportional to $\sqrt{N_h}$
Result: Similar RSE across provinces while maintaining national efficiency

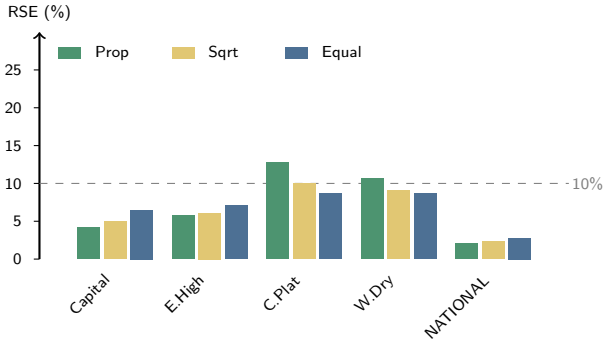
Zambara: Three Allocation Methods Compared

Sample Allocation ($n = 20,000$) by Method

Province	Pop %	Prop	Sqrt	Equal
Capital Region	26.8	5,360	3,820	2,222
Eastern Highlands	16.7	3,340	3,010	2,222
Southern Cape	10.7	2,140	2,410	2,222
Coastal Plains	9.9	1,980	2,320	2,222
Northern Bushveld	9.0	1,800	2,210	2,222
Mining Belt	6.7	1,340	1,910	2,222
Eastern Forests	7.6	1,520	2,030	2,222
Central Plateau	5.1	1,020	1,670	2,222
Western Drylands	7.4	1,480	2,000	2,222
Total	**100**	**20,000**	**20,380**	**20,000**

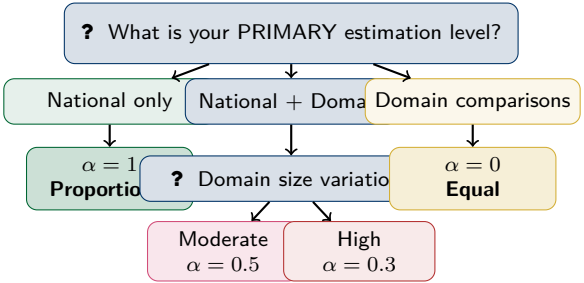
Square root shifts 1,500 interviews from Capital to smaller provinces.

Precision Impact: RSE by Province and Method



Square root brings C.Plateau from 12.8% to 10.0% RSE while keeping national RSE acceptable (2.3%).

Choosing Your α: Decision Framework



Greater domain size variation (large vs small provinces) calls for lower α.

Adding Minimum Sample Constraints

Zambara Case

Lindiwe's realization: "Even with square root allocation, my smallest stratum (Central Plateau Urban) gets only 340 interviews. That's too few for reliable estimates. I need a **minimum constraint**."

Constrained Allocation

For each stratum h, ensure $n_h \geq n_{min}$. If initial allocation $n_h^* < n_{min}$, set $n_h = n_{min}$ and reallocate remaining sample proportionally among unconstrained strata.

Calculate initial n_h^* → Apply minimum: $n_h = \max(n_h^*, n_{min})$ → Redistribute excess to large

Common n_{min} values: 300–500 for province, 100–200 for sub-domain

Zambara: Square Root with Minimum Constraint

$n = 20,000$, $\alpha = 0.5$, $n_{min} = 400$ **per province**

Province	**Initial n_h^***	**Below Min?**	**Final n_h**	**Weight**
Capital Region	3,820	No	3,720	0.72
Eastern Highlands	3,010	No	2,930	0.93
Southern Cape	2,410	No	2,350	1.00
Coastal Plains	2,320	No	2,260	1.03
Northern Bushveld	2,210	No	2,150	1.09
Mining Belt	1,910	No	1,860	1.26
Eastern Forests	2,030	No	1,980	1.17
Central Plateau	1,670	No	1,630	1.44
Western Drylands	2,000	No	1,950	1.21
Total	**20,380**	—	**20,000**	—

No province fell below minimum. Weights range from 0.72 to 1.44 (modest variation).

Understanding the Weight Implications

Weight Calculation

Base weight = inverse of selection probability:

$$w_h = \frac{N_h}{n_h}$$

Capital: $w = \frac{5{,}120{,}000}{3{,}720} = 1,376$

C.Plateau: $w = \frac{980{,}000}{1{,}630} = 601$

Ratio: $1,376/601 = 2.3$

Weight Variation

Rule of thumb:
Weight CV < 1.0 is acceptable
Weight ratio < 5 is manageable

Zambara:
Weight ratio = 2.3 (acceptable)
Weight CV = 0.21 (good)

High weight variation increases variance of weighted estimates.

Key Insight

Trade-off: More aggressive reallocation (lower α) improves domain precision but increases weight variation and harms national precision.

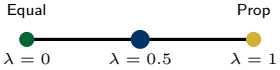
Advanced: Composite Allocation

Composite Allocation Formula

Blend proportional and equal allocation with weight λ:

$$n_h = n \times \left[\lambda \times \frac{N_h}{N} + (1-\lambda) \times \frac{1}{H}\right]$$

where $\lambda = 1$ gives proportional, $\lambda = 0$ gives equal



Linear blend between two extremes.

Comparison with Power:

- Power (α): Non-linear, smoother
- Composite (λ): Linear, simpler
- Both achieve similar goals
- Power more common in practice

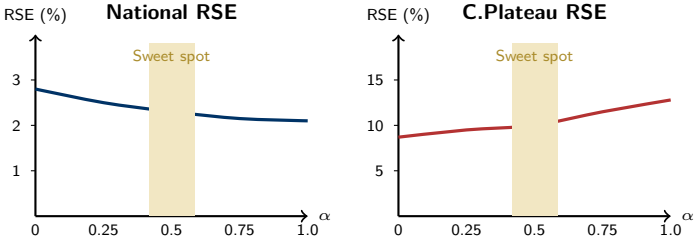
R Code: Power Allocation

```r
# Power allocation function
power_allocation <- function(N_h, n_total, alpha = 0.5, n_min = NULL) {
  # Initial allocation
  N_power <- N_h^alpha
  n_h <- n_total * N_power / sum(N_power)

  # Apply minimum constraint if specified
  if (!is.null(n_min)) {
    below_min <- n_h < n_min
    n_h[below_min] <- n_min
    remaining <- n_total - sum(n_h[below_min])
    n_h[!below_min] <- remaining * N_power[!below_min] /
                       sum(N_power[!below_min])
  }
  return(round(n_h))
}

# Apply to Zambara
strata$n_sqrt <- power_allocation(strata$N_h, 20000, alpha = 0.5)
strata$n_power3 <- power_allocation(strata$N_h, 20000, alpha = 0.3)
strata$n_const <- power_allocation(strata$N_h, 20000, alpha = 0.5,
                                   n_min = 400)
```

STATA Code: Power Allocation

```
* Power allocation program
program define power_alloc
    args alpha n_total

    * Calculate N^alpha
    gen N_power = N_h^`alpha'
    egen N_power_sum = total(N_power)

    * Initial allocation
    gen n_alloc = round(`n_total' * N_power / N_power_sum)

    * Adjust for rounding
    egen n_sum = total(n_alloc)
    replace n_alloc = n_alloc + (`n_total' - n_sum) if _n == 1

    drop N_power N_power_sum n_sum
end

* Apply different alpha values
power_alloc 1.0 20000    // Proportional
rename n_alloc n_prop
power_alloc 0.5 20000    // Square root
rename n_alloc n_sqrt
power_alloc 0.3 20000    // More aggressive
rename n_alloc n_power3
```

Sensitivity: How Much Does α Matter?



$\alpha = 0.5$ is often the "sweet spot" — modest increase in national RSE for substantial domain improvement.

Power Allocation in Practice

Allocation Approaches Used by Major Surveys

Survey	Allocation	Rationale
SA GHS	Near-proportional	Multi-purpose, provincial domains
SA LFS	Proportional + min	Employment by province required
DHS	Power ($\alpha \approx 0.5$)	Regional health estimates
LSMS	Proportional	Poverty at national level
EU-SILC	Minimum + prop	EU requires country precision
US CPS	Proportional + min	State unemployment estimates

Key Insight

Pattern: Most large-scale surveys use proportional as baseline with minimum constraints or power adjustment when domain estimates are critical.

Beyond Variance: Cost-Optimal Allocation

Cost-Optimal Allocation (Cochran)

When travel costs vary by stratum:

$$n_h = n \times \frac{N_h S_h / \sqrt{c_h}}{\sum_{k=1}^{H} N_k S_k / \sqrt{c_k}}$$

where c_h = cost per interview in stratum h

Implication:

Over-sample cheap strata (urban, accessible)
Under-sample expensive strata (remote, rural)

Zambara example:

Western Drylands: $c_h = 2.5\times$ urban cost
$\rightarrow$ Reduce sample by $1/\sqrt{2.5} = 0.63$

Trade-off: Cost savings vs. domain precision

Source: Cochran (1977), Section 5A.7

Cost-optimal allocation covered in detail in Lab 1.4.

Lindiwe's Final Allocation Design

Zambara Case

After careful analysis, Lindiwe finalizes: "I'll use **power allocation with** $\alpha = 0.6$ and a **minimum of 800 HH per province**. This gives me provincial RSEs under 12% while keeping national RSE at 2.2%. Weight variation is modest (CV = 0.18)."

ZAMBARA HLC SURVEY — FINAL ALLOCATION DESIGN

- ✔ Method: Power ($\alpha = 0.6$)
- ✔ Total sample: $n = 20,000$ HH
- ✔ Minimum: 800 HH/province
- ✔ National RSE: 2.2%
- ✔ Max provincial RSE: 11.5%
- ✔ Weight CV: 0.18

Design balances national precision, provincial estimates, and operational feasibility.

Allocation Strategies: Complete Summary

Proportional	Power	Equal
$n_h \propto N_h$	$n_h \propto N_h^\alpha$	$n_h = n/H$
Best nation	Balanced	Best domain

$\alpha = 1$ $\alpha = 0.5$ $\alpha = 0$

Recommendation: Power ($\alpha = 0.5$–0.7) with minimum constraints for multi-level surveys

Lab 1.4: Compute allocations for your country

Key formula: $n_h = n \times \frac{N_h^\alpha}{\sum N_k^\alpha}$

Next: Module 1.6 — Handling Imperfect Frames

MODULE 1.6

The Imperfect Frame

When Reality Doesn't Match the Map

Ghost Villages ⇄ Boundary Changes ⚠ Frame Errors

Lindiwe's Troubling Discovery

Zambara Case

Three weeks before fieldwork begins... Lindiwe runs a quality check on the sampling frame. The results are alarming: "94 PSUs have problems—some appear in multiple provinces, others have codes that don't match any administrative boundary. How did this happen? And what do I do about it?"

FRAME QUALITY ALERT

Total PSUs checked: **3,455**

PSUs with problems: **94 (2.72%)**

Affected households: **198 (0.89%)**

Source: Lab 1.5 Frame Mismatch Analysis

What is a Sampling Frame?

Sampling Frame

A **sampling frame** is a list (or device) that identifies and locates all units in the target population from which a sample can be drawn.

✔ Ideal Frame Properties

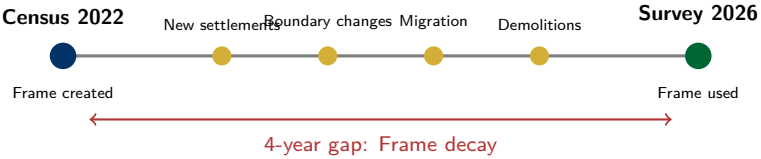
- Complete (no missing units)
- Accurate (correct information)
- Current (up-to-date)
- Non-duplicated (each unit once)
- Accessible (can locate units)

✖ Real-World Problems

- Undercoverage (missing areas)
- Overcoverage (non-existent units)
- Duplicates (same unit twice)
- Outdated boundaries
- Incorrect classifications

Source: Groves et al. (2009), Survey Methodology, Chapter 3

The Census-Based Sampling Frame



Key Insight

Frame decay: The longer since the census, the more outdated the frame. Zambara's frame is 4 years old—new informal settlements have emerged, some villages have been abandoned, and administrative boundaries have changed.

Types of Frame Errors

UNDERCOVERAGE
Units in population
NOT in frame
("Ghost villages")
New settlements, nomads

OVERCOVERAGE
Units in frame
NOT in population
("Phantom PSUs")
Demolished areas, merged

DUPLICATES
Same unit appears
multiple times
(Selection bias)
Coding errors, mergers

MISCLASSIFICATION
Unit in wrong stratum
or domain
(Stratification error)
Boundary changes

Each error type requires different remediation strategies.

Zambara Frame Analysis: Problem Overview

Total PSUs checked	PSUs with problems	Households affected
3,455	**94 (2.7%)**	**198 (0.9%)**

Problems by Severity:

CRITICAL: 50 | HIGH: 18 | MEDIUM: 34

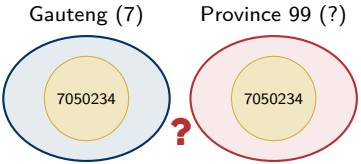
Caution

50 CRITICAL issues require immediate attention before fieldwork. These are duplicate PSU codes appearing in multiple provinces—a data integrity crisis.

Problem Type 1: Duplicate PSUs (CRITICAL)

Zambara Case

Lindiwe investigates: "PSU code 7050234 appears in BOTH Gauteng AND an invalid province 99. How is this possible? If selected, which location do we visit?"



Same code, different locations!

Impact

Found: 50 PSUs with duplicates

Causes:

- Data entry errors
- Province code corruption
- Frame merge problems

Impact:

- Double selection probability
- Cannot determine location

Problem Type 2: Malformed PSU Codes (MEDIUM)

Expected PSU Format

Standard format: PDDNNNN (7–8 digits)

P	Province code (1–9)
DD	District code (01–99)
NNNN	Sequential number

Example: 7050234
= Province 7 (Gauteng), District 05, PSU 0234

Found Problems

34 malformed codes:

X1234567 — Non-numeric
12345 — Too short
3050234 in Province 7 — Prefix mismatch

Cannot match to geography!

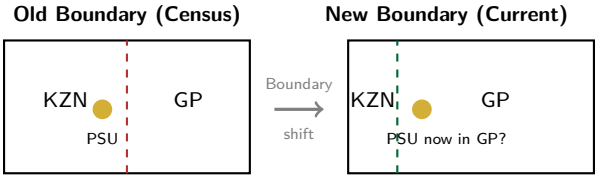
Key Insight

Fix: Create a crosswalk table mapping malformed codes to correct codes. Verify against original census frame documentation.

Problem Type 3: Province-District Mismatch (HIGH)

Zambara Case

The boundary change problem: "PSU 5020246 is coded in KwaZulu-Natal (Province 5), but its district code starts with 7 (Gauteng). Was there a boundary change, or is this a coding error?"



Found: 18 PSUs with province-district inconsistencies. May reflect boundary demarcation changes.

Problem Type 4: Invalid Province Codes (HIGH)

Valid Province Codes

Code	Province
1	Western Cape
2	Eastern Cape
3	Northern Cape
4	Free State
5	KwaZulu-Natal
6	North West
7	Gauteng
8	Mpumalanga
9	Limpopo

Found Invalid

4 PSUs with code 99

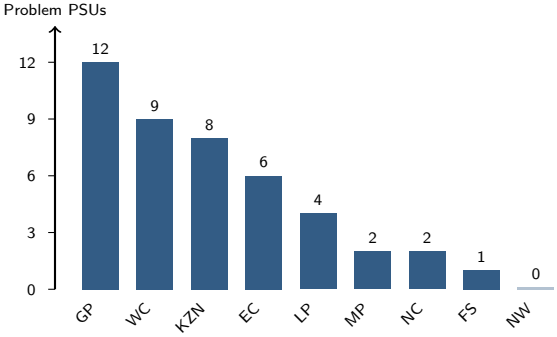
Province "99" does not exist!

Possible causes:

- Missing data placeholder
- Import/export corruption
- Manual coding error

Action: Trace to original frame or exclude from sample.

Where Are the Problems? Geographic Distribution



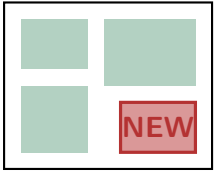
Gauteng (GP) has the most problems (12 PSUs)—likely due to rapid urbanization and complex metro boundaries. North West has no detected problems.

Ghost Villages: The Undercoverage Problem

Zambara Case

Field team report: "We found three new informal settlements near Johannesburg that don't appear on our frame. Hundreds of households live there, but we can't interview them because they weren't in the census."

Frame Coverage



In frame Missing

Undercoverage Impact

Who is missed:

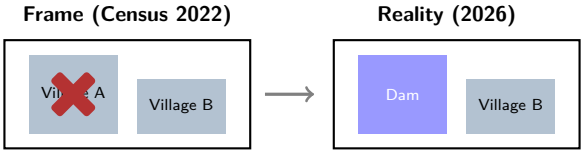
- New informal settlements
- Peri-urban expansion
- Nomadic populations
- Homeless populations

Bias direction:
Estimates may miss poorest, most vulnerable groups.

Phantom PSUs: The Overcoverage Problem

Zambara Case

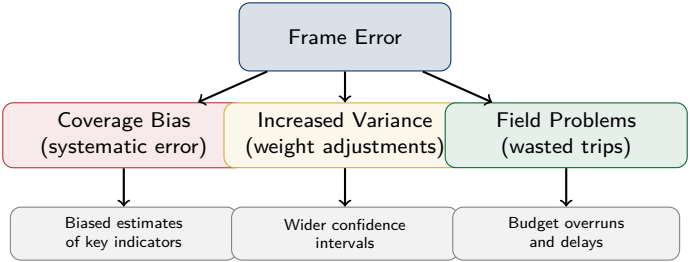
Another field team report: "PSU 2150364 is supposed to be in Eastern Cape, but when we arrived, there was nothing there—just empty land. The village was relocated three years ago for a dam project."



Key Insight

Solution: When field team cannot locate PSU, document the situation, mark as "non-existent," and do NOT substitute. Adjust weights for overcoverage.

Impact of Frame Errors on Survey Quality



Zambara impact: 2.7% of PSUs affected means potential 2–3% coverage bias if problem PSUs are systematically different from clean PSUs.

Remediation Strategy: Lindiwe's Action Plan

Week 1	Week 2	Week 3
CRITICAL Resolve duplicates (50 PSUs) Assign unique codes	**HIGH** Fix province codes (22 PSUs) Verify boundaries	**MEDIUM** Correct malformed (34 PSUs) Create crosswalk

✔ Clean frame ready for sampling ✔

All 94 problem PSUs must be resolved BEFORE sample selection to ensure valid selection probabilities.

Resolving Duplicate PSU Codes

Step-by-Step Resolution

Step 1: Identify all duplicates in the frame
Step 2: For each duplicate, trace to original census documentation
Step 3: Determine which record is correct based on:

- Geographic coordinates (if available)
- District/municipality assignment
- EA (Enumeration Area) nesting

Step 4: Assign new unique codes OR delete erroneous records
Step 5: Document all changes in a frame modification log

Caution

Never delete without documentation. Every frame modification must be logged for audit trail and to enable correct weighting.

Handling Boundary Changes

When Boundaries Changed

Option A: Update to current

- Reclassify PSU to new province
- Update stratum assignment
- Recalculate probabilities

Option B: Retain census codes

- Keep original codes
- Note discrepancy in metadata
- Weight to census totals

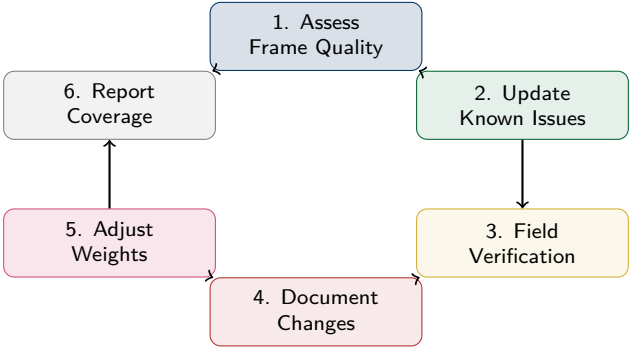
Recommendation

For stratification: Use current boundaries (administrative relevance)
For weighting: Use census totals (known population counts)
Document both:

- Census-based classification
- Current administrative unit
- Date of boundary change

Frame Maintenance Best Practices

Continuous Frame Improvement Cycle



Apply this cycle before each survey round. GIS and satellite imagery help identify new settlements.

Imperfect Frames: Key Takeaways

No frame is perfect—pla

Duplicates are CRITICAL (50 found)

Boundary changes cause m

Malformed codes prevent matching (34)

Resolve ALL frame issues BEFORE sample selection

Zambara Case

Lindiwe's lesson: "I almost started fieldwork with 94 problem PSUs. The frame quality check saved us from potential disaster. Now I run Lab 1.5 before EVERY survey round."

Next: Module 1.7—Surveying Rare and Hard-to-Reach Populations

Lab Exercise 1.5: Frame Quality Assessment

Hands-On Exercise (45 minutes)

Objective: Run the frame mismatch analysis on your country's sampling frame

Steps:

1. Load the R script: `lab1_5_imperfect_frames_R.R`
2. Set file paths to your survey data and shapefiles
3. Run the analysis and examine output files
4. Review the problem report and severity classifications
5. Discuss: What remediation steps would you take?

Discussion questions:

- What frame errors are most common in your country?
- How do you currently handle boundary changes?
- What is your frame update frequency?

Output files: `frame_mismatch_report.csv`, `frame_quality_summary.csv`, visualization PNGs

MODULE 1.6 CONTINUED

Frame Correction Techniques

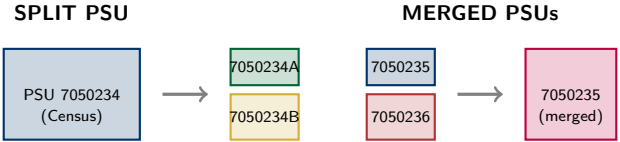
Split PSUs, Merged PSUs, and GIS Validation

Splits Merges Shapefile Overlay Procedures

The PSU Boundary Challenge

Zambara Case

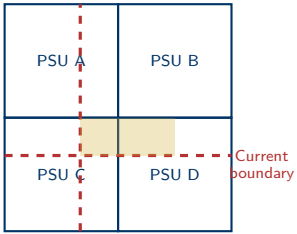
Lindiwe's next challenge: "Even after fixing the code errors, we have a bigger problem. Some PSUs have physically changed since the census—some were split into multiple areas, others were merged. How do I handle selection probabilities when the units themselves have changed?"



Both scenarios affect selection probabilities and require specific correction procedures.

GIS Shapefile Overlay: The Detection Method

Layer 1: Census PSU Boundaries



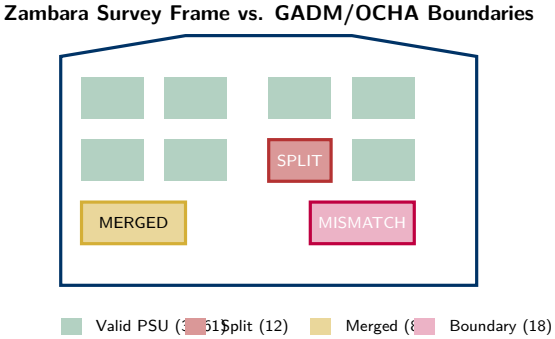
Yellow areas: Boundary mismatches detected

Overlay Process

Step 1: Load census PSU shapefile
Step 2: Load current admin boundaries (GADM/OCHA)
Step 3: Spatial join on geometry
Step 4: Flag non-matching polygons
Step 5: Calculate area differences

Tools: R (sf), STATA (spmap), QGIS, ArcGIS

Lab 1.5: GIS Validation Results



Of 94 problem PSUs: 12 splits, 8 merges, 18 boundary mismatches, 50 duplicates, 4 invalid codes.

Understanding Split PSUs

Split PSU Definition

A **split PSU** occurs when a single census enumeration area has been subdivided into two or more distinct geographic units since the frame was created.

? Why Splits Occur

- Rapid population growth
- New road/infrastructure dividing area
- Administrative redistricting
- Urban densification
- New municipal boundaries

⚠ Impact on Sampling

- Original MOS outdated
- Selection probability unclear
- Which segment to visit?
- Listing becomes ambiguous
- Weighting complexity

Zambara finding: 12 PSUs have been split since Census 2022, primarily in Gauteng (urban growth).

Split PSU: Visual Example

Census 2022

PSU 7050234
MOS = 450 HH
Area = 2.4 km^2

New highway constructed (2024) →

Survey 2026

Segment A
MOS = 280 HH

HIGHWAY N1

Segment B
MOS = 170 HH

? Which segment do we sample?

Procedure: Handling Split PSUs

Split PSU Correction Protocol

Option A: Random Segment Selection (Recommended)

1. Identify all segments created from original PSU
2. Assign selection probability proportional to segment MOS
3. Randomly select ONE segment using PPS
4. Document selection and calculate adjusted weight

Option B: Take-All Approach

1. Treat all segments as single PSU
2. List households in ALL segments
3. Sample from combined list
4. Weight using original PSU probability

Key Insight

Split PSU: Weight Calculation Example

Given Information

Original PSU 7050234:

- Census MOS = 450 HH
- Original weight = 125
- Selected with $\pi = 0.008$

After split:

- Segment A: MOS = 280 HH
- Segment B: MOS = 170 HH
- Segment A randomly selected

Adjusted Weight

Step 1: Segment selection prob.

$$P(A) = \frac{280}{450} = 0.622$$

Step 2: Adjusted weight

$$w_A = 125 \times \frac{450}{280} = 201$$

Interpretation:
Each HH in Segment A now represents 201 HH in population (vs. 125 originally).

Understanding Merged PSUs

Merged PSU Definition

A **merged PSU** occurs when two or more census enumeration areas have been combined into a single administrative or geographic unit since the frame was created.

? Why Merges Occur

- Population decline
- Administrative efficiency
- Municipal consolidation
- Village abandonment
- Urban renewal projects

⚠ Impact on Sampling

- Multiple selection if both selected
- Combined MOS different from sum
- Original boundaries lost
- Double-counting risk
- Stratification disrupted

Zambara finding: 8 PSU pairs have been merged, primarily in Northern Cape (population decline).

Merged PSU: Visual Example

Census 2022

PSU 3060101
MOS = 85 HH

PSU 3060102
MOS = 62 HH

Municipal merger (2023) →

Survey 2026

PSU 3060101
Combined MOS = 142 HH
(3060102 no longer exists)

⚠ What if BOTH were selected?

Procedure: Handling Merged PSUs

Merged PSU Correction Protocol

Scenario 1: Only ONE original PSU was selected

- Proceed with merged area as single PSU
- List all households in combined area
- Adjust weight: $w_{merged} = w_{selected} \times \frac{MOS_{selected}}{MOS_{merged}}$

Scenario 2: BOTH original PSUs were selected

- Treat as single selection (avoid double-counting)
- Calculate composite weight: $w_{composite} = \frac{1}{\pi_A + \pi_B - \pi_A \pi_B}$
- List households once only

Caution

Critical: Always check if multiple selected PSUs have merged before fieldwork. Double visits waste resources and bias estimates.

Merged PSU: Weight Calculation Example

Both PSUs Selected

PSU 3060101:

- Census MOS = 85 HH
- $\pi_A = 0.012$
- $w_A = 83.3$

PSU 3060102:

- Census MOS = 62 HH
- $\pi_B = 0.009$
- $w_B = 111.1$

Merged area: MOS = 142 HH

Composite Weight

Step 1: Combined selection prob.

$$\pi_{AB} = \pi_A + \pi_B - \pi_A \pi_B$$

$$= 0.012 + 0.009 - 0.000108$$

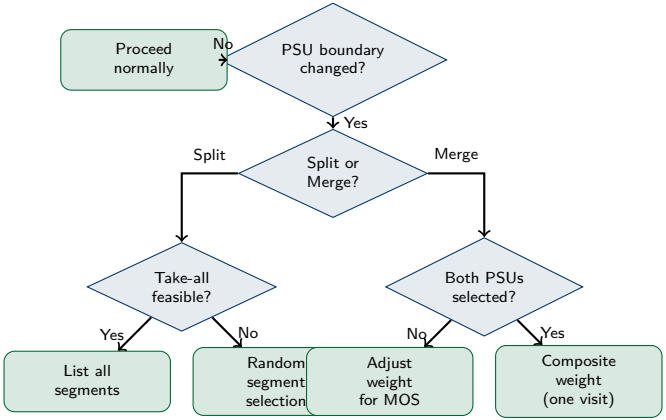
$$= 0.0209$$

Step 2: Composite weight

$$w_{merged} = \frac{1}{0.0209} = 47.8$$

Note: Lower than either original because area had higher combined selection probability.

Decision Tree: Handling PSU Boundary Changes



R Code: GIS Shapefile Overlay

```r
# Load required packages
library(sf)
library(dplyr)

# Load shapefiles
census_psu <- st_read("census_psu_boundaries.shp")
current_admin <- st_read("gadm_admin2.shp")

# Spatial join to detect mismatches
overlay <- st_join(census_psu, current_admin, left = TRUE)

# Identify PSUs crossing current boundaries
boundary_issues <- overlay %>%
  group_by(psu_code) %>%
  filter(n_distinct(admin2_code) > 1) %>%
  summarise(
    n_admin_units = n_distinct(admin2_code),
    admin_units = paste(unique(admin2_code), collapse = ", ")
  )

# Calculate area differences (potential splits/merges)
```

STATA Code: Frame Boundary Validation

```
* Convert shapefiles to STATA format
shp2dta using census_psu_boundaries, ///
    database(psu_db) coordinates(psu_xy) replace

shp2dta using gadm_admin2, ///
    database(admin_db) coordinates(admin_xy) replace

* Load and merge
use psu_db, clear
merge m:1 admin2_code using admin_db, keep(master match)

* Flag boundary mismatches
bysort psu_code: egen n_admin = nvals(admin2_code)
gen boundary_issue = (n_admin > 1)

* List problem PSUs
list psu_code prov district n_admin if boundary_issue == 1

* Export for remediation
export delimited psu_code prov district n_admin ///
    using "boundary_issues.csv" if boundary_issue == 1, replace
```

Frame Modification Log: Documentation Standard

Required Fields for Every Frame Change

PSU Code	Change Type	Date	Old Value	New Value	Justification
7050234	SPLIT	2026-02-15	1 PSU	2 segments	Highway N1
3060101	MERGE	2026-02-16	2 PSUs	1 PSU	Municipal
5020246	RECODE	2026-02-17	Prov 5	Prov 7	Boundary 2016
X1234567	DELETE	2026-02-18	Invalid	Removed	Duplicate

Key Insight

Golden rule: Every frame modification must be logged with who, what, when, and why. This enables audit and correct weight calculation.

Quality Metrics: Before and After Correction

BEFORE Correction

Total PSUs: 3,455
Problem PSUs: 94 (2.72%)
- Duplicates: 50
- Boundary: 18
- Splits/Merges: 20

Correction Protocol →

AFTER Correction

Total PSUs: 3,449
Problem PSUs: 0 (0%)
- Resolved: 88
- Deleted: 6
- Documented: 94

Frame coverage: 99.8% | Boundary match: 100% | Ready for sampling: ✔

Common Pitfalls in Frame Correction

✖ DON'T

- Delete PSUs without documentation
- Substitute another PSU for missing one
- Ignore split segments (visit just one)
- Visit merged PSUs twice
- Forget to adjust weights
- Make corrections during fieldwork

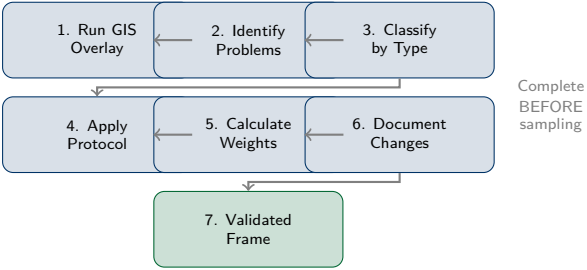
✔ DO

- Log every modification
- Use random segment selection
- Calculate composite weights
- Verify with GIS overlay
- Complete corrections pre-fieldwork
- Train field teams on changes

⚠ Caution

Never substitute: If PSU cannot be visited, document as non-response. Substitution introduces unknown bias and invalidates probability calculations.

Frame Correction: Complete Workflow Summary



Zambara Case

Lindiwe's final status: "We resolved all 94 frame issues in 3 weeks. The modification log has 94 entries with full documentation. Our frame is now 100% validated against current GADM boundaries. We're ready to draw the sample."

Reference Card: Frame Correction Formulas

Split PSU Formulas

Segment selection probability:

$$P(segment) = \frac{MOS_{segment}}{MOS_{original}}$$

Adjusted weight:

$$w_{segment} = w_{original} \times \frac{MOS_{original}}{MOS_{segment}}$$

Validation check:

$$\sum_s MOS_s = MOS_{original}$$

Merged PSU Formulas

Combined selection prob:

$$\pi_{merged} = \pi_A + \pi_B - \pi_A \pi_B$$

Composite weight (both selected):

$$w_{merged} = \frac{1}{\pi_A + \pi_B - \pi_A \pi_B}$$

Single selection weight:

$$w_{merged} = w_{selected} \times \frac{MOS_{selected}}{MOS_{merged}}$$

Source: Lohr (2022), Sampling: Design and Analysis, Chapter 6

MODULE 1.6 CONTINUED

Advanced Frame Maintenance

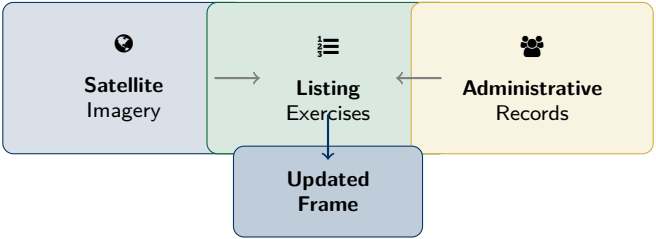
Satellite Imagery and Listing Exercises

Remote Sensing | Listing | Updates | Checklists

The Frame Maintenance Challenge

Zambara Case

Lindiwe looks ahead: "We fixed all 94 frame problems for this survey. But what about next year? The frame will continue to decay. We need a sustainable maintenance strategy that doesn't require waiting for the next census."



Satellite Imagery for Frame Updates

Remote Sensing for Sampling Frames

Satellite imagery provides visual evidence of settlement changes between censuses, enabling detection of new construction, demolition, and population shifts without costly field visits.

✔ What Imagery Can Detect

- New residential construction
- Informal settlement growth
- Building demolitions
- Road network changes
- Agricultural-to-urban conversion
- Flood/disaster damage

✖ Limitations

- Cannot count households
- Building dwelling unit
- Cloud cover issues
- Resolution constraints
- Requires technical expertise
- Cost of high-res imagery

Checklist: Satellite Imagery Sources

Available Imagery Sources for NSOs

	Source	Resolution	Cost	Coverage	Best For
☑	Sentinel-2	10m	Free	Global	Large areas
☑	Landsat 8/9	30m	Free	Global	Historical
☑	Google Earth	Variable	Free	Urban	Visual check
☐	Planet	3-5m	Paid	Daily	Change det.
☐	Maxar/DG	0.3-0.5m	Expensive	On-demand	Building cnt

Recommendation: Start with free Sentinel-2 for broad screening, then use higher resolution for priority areas.

Checklist: Satellite Change Detection Workflow

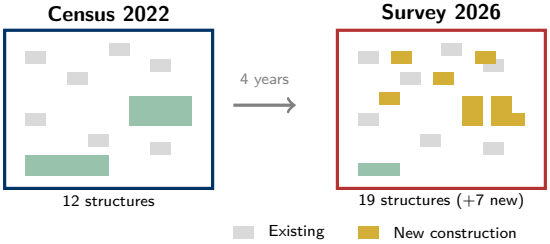
Step-by-Step Change Detection

- ☐ **Acquire baseline imagery** (census date or closest available)
- ☐ **Acquire current imagery** (within 6 months of survey)
- ☐ **Co-register images** (align to same coordinate system)
- ☐ **Calculate change indices** (NDBI, NDVI difference)
- ☐ **Apply threshold** to identify significant changes
- ☐ **Overlay with PSU boundaries** from sampling frame
- ☐ **Flag PSUs** with detected changes for field verification
- ☐ **Prioritize** based on change magnitude

Key Insight

Key metric: Normalized Difference Built-up Index (NDBI) increases indicate new construction; decreases may indicate demolition.

Visual Comparison: Before and After



PSU shows 58% increase in structures. **Action:** Flag for listing update, potential MOS revision.

Emerging Technology: AI Building Detection

How It Works

Machine Learning models:

- Trained on labeled building footprints
- Detect structures in new imagery
- Count buildings per PSU
- Estimate dwelling units

Available tools:

- Microsoft Building Footprints
- Google Open Buildings
- Facebook Population Maps

Caveats

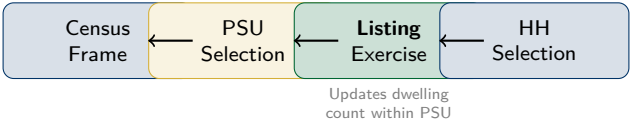
- Building $\neq$ household
- Multi-unit structures
- Non-residential buildings
- Model accuracy varies
- Rural areas less accurate
- Still requires validation

Best use:
Relative change detection, not absolute counts.

Listing Exercises: Definition and Purpose

Listing Exercise

A **listing exercise** is a systematic enumeration of all dwelling units within selected PSUs, conducted prior to household sampling to create an up-to-date frame for second-stage selection.



Key benefit: Creates fresh second-stage frame, eliminating undercoverage of new dwellings within selected PSUs.

Checklist: Pre-Listing Preparation

Before Deploying Listers

Materials Preparation:

- ☐ PSU boundary maps (hardcopy and digital)
- ☐ Listing forms (paper or tablet)
- ☐ GPS devices or smartphone with GPS app
- ☐ Unique dwelling unit ID system
- ☐ Reference photos of PSU landmarks

Training Requirements:

- ☐ Dwelling unit definition (what counts?)
- ☐ Boundary identification (where to stop?)
- ☐ Systematic coverage pattern (route planning)
- ☐ Quality control procedures
- ☐ Safety protocols

Checklist: Listing Form Required Elements

Essential Fields

- ☑ PSU identification code
- ☑ Sequential dwelling number
- ☑ Structure type (house, flat, etc.)
- ☑ Number of households
- ☑ GPS coordinates
- ☑ Physical address/description
- ☑ Lister ID and date

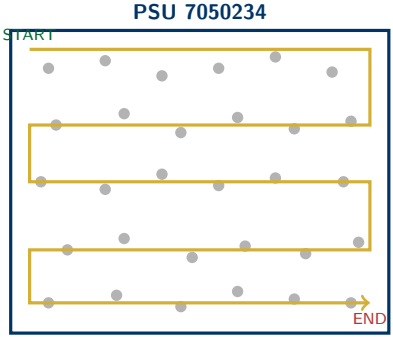
Optional Fields

- ☐ Building material
- ☐ Estimated household size
- ☐ Occupancy status
- ☐ Contact name/phone
- ☐ Accessibility notes
- ☐ Photo of dwelling
- ☐ Business presence

Caution

Critical: Every dwelling must have a unique ID. Use format: PSU-XXXX where XXXX is sequential within PSU.

Listing Pattern: Systematic Coverage



Total listed: 30 dwellings

Serpentine pattern: Ensures complete coverage without backtracking. Start at consistent corner (e.g., NW).

Checklist: Listing Quality Control

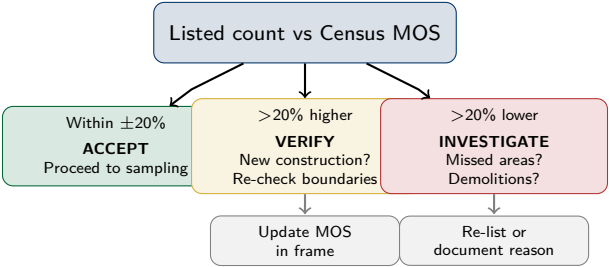
During Listing (Field Supervisor)

- ☐ Verify PSU boundaries are correctly identified
- ☐ Spot-check 10% of listed dwellings exist
- ☐ Ensure sequential numbering has no gaps
- ☐ Confirm GPS coordinates are within PSU
- ☐ Check dwelling count against census MOS ($\pm$20%)
- ☐ Review for duplicate entries

After Listing (Office)

- ☐ Data entry verification (if paper forms)
- ☐ GPS coordinates plot within PSU boundary
- ☐ Flag PSUs with $>$30% deviation from expected
- ☐ Re-list if serious problems detected

Handling Listing Discrepancies



Always document the reason for discrepancies. This information is valuable for future frame updates.

Modern Listing: CAPI/CAWI Integration

CAPI Listing Benefits

- ☑ Automatic GPS capture
- ☑ Built-in validation rules
- ☑ Photo documentation
- ☑ Real-time data upload
- ☑ No transcription errors
- ☑ Immediate QC dashboard

Software Options

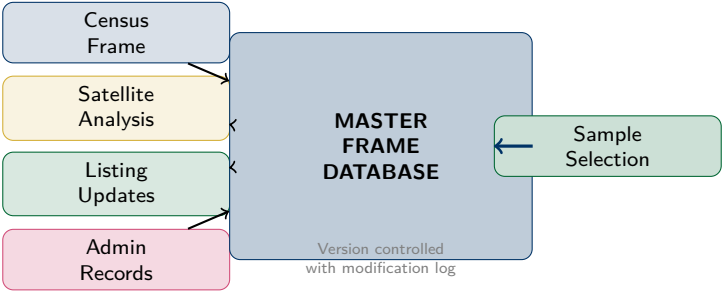
- Survey Solutions (World Bank)
- CSPro (US Census Bureau)
- ODK/KoboToolbox
- SurveyCTO
- Fulcrum

Key feature:
Offline capability for rural areas

Key Insight

Best practice: Link listing data directly to sampling software. Automatic household selection from fresh listing eliminates manual errors.

Integrating Updates into the Master Frame



Checklist: Frame Update Schedule

Recommended Update Frequency by Source

Update Type	Frequency	Trigger	Responsibility
Satellite screening	Annual	Pre-survey	GIS unit
Listing exercise	Per survey	PSU selection	Field ops
Admin boundary	As changed	Gazette notice	Geography
Population estimates	Annual	Mid-year	Demography
Full frame rebuild	10 years	Post-census	All units

Frame updates should be **cumulative**—each update builds on previous version, never starts from scratch.

Checklist: Frame Update Documentation

Required Documentation for Each Update

- ☐ **Version number** and date
- ☐ **Source** of update (satellite, listing, admin)
- ☐ **Geographic scope** (national, regional, PSU-level)
- ☐ **Changes summary** (PSUs added, deleted, modified)
- ☐ **Quality assessment** (coverage rate, validation results)
- ☐ **Responsible person** and approval
- ☐ **Backup location** of previous version

Good: "Frame v3.2, March 2026, satellite update of Gauteng urban PSUs, +23 new PSUs identified"

Bad: "Updated frame.xlsx"

Zambara: Annual Frame Maintenance Plan

Zambara Case

Lindiwe's sustainable strategy: "I've designed a 12-month maintenance cycle. We'll never again have 94 problems accumulate before discovery."

Satellite screening	Field verification	Frame update	Quality report
Q1	Q2	Q3	Q4

Ready-to-use frame for annual surveys

Advanced Frame Maintenance: Key Takeaways

- ☑ Satellite imagery detects large-scale changes between censuses
- ☑ Listing exercises update dwelling counts in selected PSUs
- ☑ CAPI integration improves listing quality and efficiency
- ☑ Annual maintenance prevents problem accumulation
- ☑ Documentation and version control are essential

Key Insight

Golden rule: A well-maintained frame is a continuous asset, not a one-time product. Invest in maintenance to protect your sampling investment.

Module 1.6 Complete: The Imperfect Frame

Module Summary: From Problem Detection to Sustainable Maintenance

Part A (181-2
Problem Detec
Ghost villag
Boundary char
94 issues fou

→

Part B (201-2
Correction Techr
Split/Merged F
Weight adjustm
GIS validatic

→

Part C (221-240)
Maintenance
Satellite imagery
Listing exercises
Annual cycle

Clean, validated, maintainable sampling frame

Zambara Case

Lindiwe's conclusion: "The frame is the foundation of every survey. We've learned to detect problems, fix them properly, and prevent future decay. Now we're ready for Module 1.7: reaching rare populations."

DAY 1 — MODULE 1.7

Surveying Rare and Hard-to-Reach Populations

When 5% Becomes a Challenge

Detection Efficiency Screening Oversampling

The Rare Population Challenge

Zambara Case

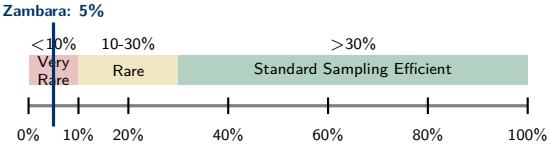
Lindiwe's next assignment: "The Ministry of Social Development wants estimates of high-income households for tax policy analysis. They need 10% precision on households earning above R58,000/month—that's only 5% of our population. My standard sample design won't work here. How do I efficiently capture a group this small?"

How do you find the
5 gold needles in 100 haystacks?

What Makes a Population "Rare"?

Rare Population

A **rare population** is a subgroup comprising a small proportion of the total population, typically less than 10-15%, making standard probability sampling inefficient for obtaining adequate sample sizes.



Examples: High-income HH, disabled persons, ethnic minorities, specific disease patients, informal workers.

The Mathematics: Why Rarity Matters

Sample Size Formula

For estimating a proportion with relative precision (CV):

$$n = \frac{z^2 \times (1 - P)}{CV^2 \times P}$$

Key insight:
As $P \to 0$, the term $\frac{1-P}{P} \to \infty$

Sample size explodes for rare populations!

Sample Size by Prevalence

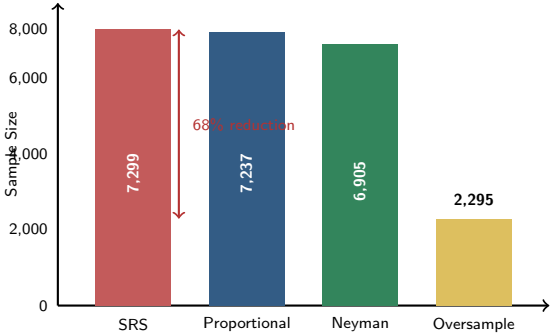
For 10% CV at 95% confidence:

Prevalence	n Required
50%	384
20%	1,537
10%	3,458
5%	**7,299**
1%	38,032

Key Insight

Zambara reality: To estimate 5% prevalence with 10% CV requires interviewing **7,299 households** to find approximately 365 high-income cases.

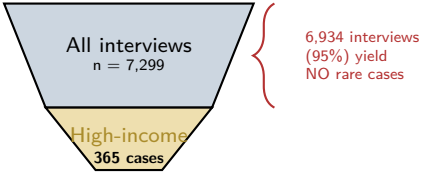
Lab 1.6 Results: The Inefficiency of SRS



Source: Lab 1.6 analysis of GHS 2024 data. Target: 10% CV for top 5% income households (P=0.05).

The Expected Yield Problem

SRS Sample: 7,299 Households



At R500/interview: R3.65M spent to find 365 high-income HH

Why Standard Sampling Fails for Rare Groups

LOW YIEL

Expected rare:
Interview 20 I
→ Find only 1
High "empty trip

HIGH COS

Cost per rare c
R500 × 20 = R1
vs. R500 for cor
20× cost multi

PRECISION PENALTY

Small rare sample
→ Wide confidence intervals
→ Unreliable subgroup analysis

Standard SRS is INEFFICIENT for rare populations

Two Strategic Approaches

Rare Population Challenge

SCREENING

Use cheap filter questions to identify rare members BEFORE full interview

- Low cost per screen
- ✔ High precision
- ✖ Requires identifier

When rare trait is observable

OVERSAMPLING

Sample more heavily from areas where rare group is concentrated

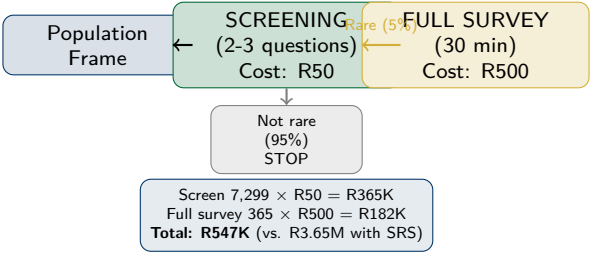
- No identifier needed
- Requires weighting
- Needs geographic proxy

When rare trait is clustered

Approach 1: Screening

Screening

Screening uses a brief, low-cost questionnaire to identify members of the rare population before administering the full (expensive) survey instrument.



Designing Effective Screening Questions

✔ Good Screening Questions

- Short (1-3 questions)
- Binary or simple categorical
- High sensitivity (catch all rare)
- Objective criteria
- Non-sensitive wording

Example (high income):
"Does your household's total monthly income exceed R50,000?"

✖ Poor Screening Questions

- Long batteries
- Complex skip patterns
- Low sensitivity (miss cases)
- Subjective assessments
- Sensitive/embarrassing

Problem example:
"Would you describe yourself as wealthy?" (subjective, sensitive)

Key Insight

Trade-off: High sensitivity catches more rare cases but increases false positives (full surveys on non-rare). Aim for 90%+ sensitivity.

Screening: Cost-Benefit Analysis

When Does Screening Save Money?

Component	SRS (No Screen)	With Screening
Screening cost	R0	7,299 × R50 = R365K
Full survey cost	7,299 × R500 = R3.65M	365 × R500 = R182K
TOTAL	**R3,650,000**	**R547,000**

Savings: R3.1M (85%)

Break-even formula: Screening is cost-effective when:

$$\frac{C_{screen}}{C_{full}} < 1 - P$$

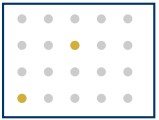
Where P = prevalence. For P=5% and C_{full}=R500: Screen cost must be < R475.

Approach 2: Disproportionate Stratified Sampling

Disproportionate Sampling

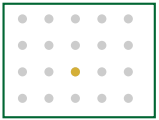
Disproportionate (over)sampling allocates more sample to strata where the rare population is concentrated, increasing the expected yield of rare cases per interview.

Urban Stratum



P = 6.5%
W = 65%

Non-Urban Stratum



P = 2.3%
W = 35%

Strategy: Oversample Urban
Allocate 50% to Urban (vs. natural 65%)
→ Higher rare case yield

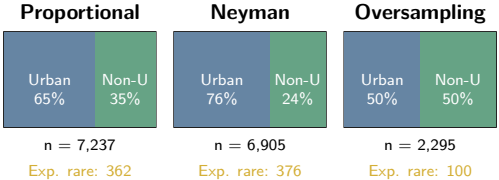
Lab 1.6: Stratum-Specific Prevalence

High-Income Household Distribution by Geographic Type

Stratum	Population	Rare HH	Prevalence	Weight
Urban	13,654	883	**6.47%**	65.2%
Non-Urban	7,286	164	2.25%	34.8%
Total	**20,940**	**1,047**	**5.00%**	**100%**

Urban prevalence (6.47%) is **2.9× higher** than Non-Urban (2.25%)
→ Urban stratum contains 84% of all high-income HH

Allocation Strategies Compared



Key Insight

Key insight: Neyman allocation automatically oversamples the high-variance (higher prevalence) stratum, improving efficiency by 5.7% over proportional.

Critical: Weighting Adjustment for Oversampling

Caution

Warning: Disproportionate sampling requires weighting adjustment. Unweighted estimates will be BIASED toward the oversampled stratum.

✖ Without Weights

50% of sample from Urban
→ Urban overrepresented
→ Estimate inflated

Biased estimate:
$\hat{P}_{unweighted} = 0.50 \times 0.0647 + 0.50 \times 0.0225$
$= 0.0436$ (4.36%)

True value: 5.00%

✔ With Weights

Weight = Population prop / Sample prop

Urban: $w_1 = 0.652/0.50 = 1.30$
Non-Urban: $w_2 = 0.348/0.50 = 0.70$

Unbiased estimate:
$\hat{P}_{weighted} = 0.652 \times 0.0647 + 0.348 \times 0.0225$
$= 0.0500$ (5.00%)

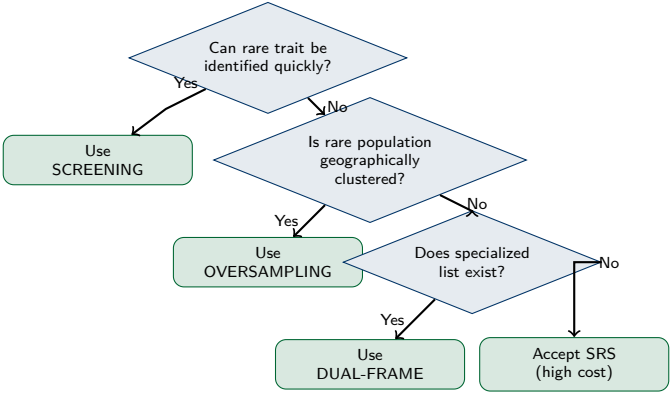
Correct!

Other Techniques for Rare Populations

MULTIPLICITY SAMPLING Ask respondents about rare trait in network members ("Do you know anyone who...")	**DUAL-FRAME SAMPLING** Combine general frame with specialized list (e.g., tax records, professional registries)
SNOWBALL/RDS Use referrals from initial rare cases to find more (for hidden populations)	**ADAPTIVE SAMPLING** Expand sample in areas where rare cases found (sequential approach)

Each technique has specific use cases. Choice depends on: existence of lists, network structure, geographic clustering, and ethical considerations.

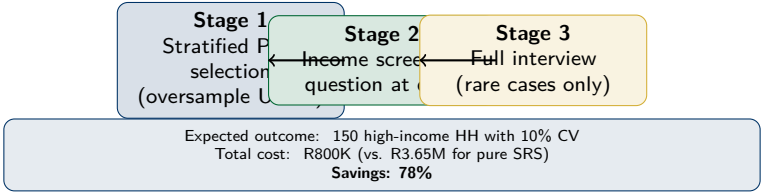
Decision Framework: Choosing the Right Approach



Zambara: Lindiwe's Recommended Strategy

Zambara Case

Lindiwe's decision: "For our high-income household survey, I recommend a two-stage approach combining screening with stratified oversampling."



Module 1.7: Key Takeaways

- Rare populations (<10%) require **specialized sampling strategies**
- SRS sample size explodes: 5% prevalence needs 7,299 HH for 10% CV
- **Screening** reduces cost by 85% when rare trait is identifiable
- **Oversampling** improves yield when rare population is clustered
- Disproportionate sampling **requires weighting** to avoid bias

Source: Kalton & Anderson (1986), "Sampling Rare Populations," *JRSS-A*; Lohr (2022), Ch. 13.

Lab 1.6: Hands-On Exercise

Lab 1.6: Sampling Rare Populations (45 minutes)

Objective: Compare sample size requirements for rare population estimation using GHS 2024 data.

Tasks:

1. Define rare population (top 5% income $\geq$ R58,012/month)
2. Calculate SRS sample size for 10% CV
3. Create Urban/Non-Urban strata and calculate prevalence
4. Compare Proportional vs. Neyman allocation
5. Design oversampling strategy for 100+ rare cases

Expected findings:

- SRS requires n = 7,299 to achieve 10% CV
- Urban stratum has 2.9$\times$ higher prevalence than Non-Urban
- Neyman allocation reduces sample by 5.4%
- Oversampling achieves 100 cases with only n = 2,295

MODULE 1.7 CONTINUED

Case Study: Zambara High-Income Survey

From Theory to Implementation

Brief Analysis Design Validation

Case Study: The Ministry Brief

Official Request from Ministry of Finance

To: Zambara National Statistics Office
From: Ministry of Finance, Tax Policy Division
Subject: Survey of High-Income Households

The Ministry requires reliable statistics on households earning above the 95th percentile (approximately R58,000/month) for:

- Progressive tax policy formulation
- Wealth distribution analysis
- Social transfer targeting refinement

Requirements:

- Precision: Coefficient of Variation $\leq 10\%$
- Minimum 100 high-income cases for subgroup analysis
- Budget constraint: R1.5 million maximum
- Timeline: 6 months from approval

Step 1: Initial Assessment

Zambara Case

Lindiwe reviews the brief: "Let me first understand what we're dealing with. I'll analyze our existing GHS data to profile the high-income population before designing the sample."

Key Questions

1. What is the exact prevalence?
2. Where are they concentrated?
3. What proxy indicators exist?
4. Can we identify them at screening?
5. What sample size is feasible?

Data Source

GHS 2024 Household File

- N = 20,940 households
- Income variable: totmhinc
- Geographic: province, geotype
- Weights available

Perfect for pilot analysis!

Step 2: Population Profiling (Lab 1.6 Results)

High-Income Population Characteristics

Total Populati	Rare Populati	Prevalence
20,940	1,047	5.00%
households	high-income H	of population

Income Threshold (95th percentile): **R58,012/month**

Source: Lab 1.6 analysis of GHS 2024 household data
Income variable: Total monthly household income (totmhinc)

Step 3: Geographic Distribution Analysis

Zambara Case

Lindiwe's discovery: "The high-income population isn't evenly distributed! Urban areas have nearly 3 times the prevalence of non-urban areas. This is our stratification opportunity."

URBAN STRATUM		NON-URBAN STRATUM
Population: 13,654 HH		Population: 7,286 HH
High-income: 883 HH		High-income: 164 HH
Prevalence: 6.47%	← 2.9× →	**Prevalence: 2.25%**
Share of frame: 65.2%		Share of frame: 34.8%
Contains 84% of all rare cases		Contains 16% of all rare cases

Step 4: Budget Reality Check

✖ SRS Approach

Sample required: 7,299 HH

Cost breakdown:

- Interviews: 7,299 × R500
- Total: **R3,649,500**

Budget: R1.5M
Shortfall: R2.15M (143%)

✖ **NOT FEASIBLE**

✔ Stratified + Screening

Sample required: 2,295 HH

Cost breakdown:

- Screening: 2,295 × R50
- Full interviews: 100 × R500
- Total: **R164,750**

Budget: R1.5M
Surplus: R1.34M

✔ **FEASIBLE**

Key Insight

Decision: Budget constraint eliminates SRS. Must use stratified oversampling with screening.

Step 5: Design Decision Matrix

Evaluating Design Options Against Requirements

Criterion	SRS	Proportional	Neyman	Oversample
CV $\leq$ 10%	✔	✔	✔	~15%
Min 100 rare	365	362	376	100
Budget $\leq$ R1.5M	✖	✖	✖	✔
Sample size	7,299	7,237	6,905	2,295
VERDICT	Reject	Reject	Reject	**SELECT**

Note: Oversampling CV of 15% exceeds target, but Ministry confirmed 100+ cases is primary requirement for subgroup analysis.

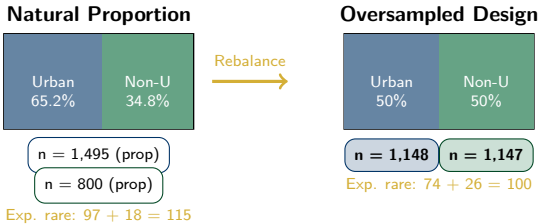
Step 6: Final Design Specification

Zambara High-Income Household Survey — Sample Design

Design: Stratified two-stage cluster sample with disproportionate allocation

Parameter	Specification
Target population	Households with income $\geq$ R58,012/month
Stratification	Urban vs. Non-Urban (geotype)
Allocation	50% Urban / 50% Non-Urban
Total sample	2,295 households
Urban allocation	1,148 households
Non-Urban allocation	1,147 households
Expected rare cases	100 (minimum requirement)
Screening question	"Does HH income exceed R50,000/month?"

Step 7: Stratum Allocation Detail



Key Insight

Trade-off: Oversampling Non-Urban (from 35% to 50%) increases representation of that stratum but requires larger weights for Urban in analysis.

Step 8: Weight Calculation Protocol

Base Weight Formula

For disproportionate sampling, base weight compensates for unequal selection probabilities:

$$w_h = \frac{W_h}{f_h} = \frac{\text{Population proportion}}{\text{Sample proportion}}$$

Stratum	Pop. Prop (W_h)	Sample Prop (f_h)	Weight (w_h)
Urban	0.652	0.500	**1.304**
Non-Urban	0.348	0.500	**0.696**

Interpretation: Each Urban household represents 1.30 population households; each Non-Urban household represents 0.70 population households. Weights sum to 1.0 across strata.

Step 9: Screening Protocol Design

Doorstep Screening Questionnaire

Introduction: "Good morning/afternoon. I am from the Zambara NSO conducting a brief survey. May I ask 2 quick questions?"

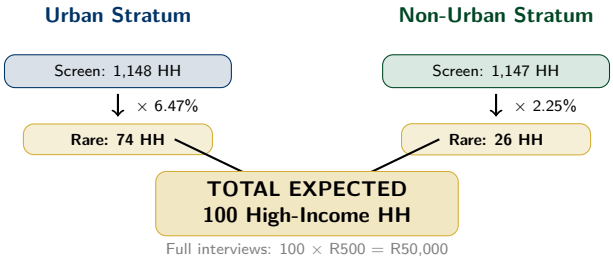
Q1. How many people usually live in this household? ____

Q2. Thinking about ALL sources of income for ALL household members, would you say your household's total monthly income is:

- ☐ Less than R30,000
- ☐ R30,000 – R49,999
- ☐ **R50,000 or more** → ELIGIBLE: Proceed to full interview
- ☐ Refused/Don't know → ELIGIBLE: Proceed (err on side of inclusion)

If NOT eligible: "Thank you for your time. Have a good day."

Step 10: Expected Outcomes Calculation



Step 11: Precision Assessment

Variance Calculation

Stratified estimator variance:

$$Var(\hat{P}_{st}) = \sum_h \frac{W_h^2 \cdot P_h \cdot Q_h}{n_h}$$

Urban:
$\frac{0.652^2 \times 0.0647 \times 0.9353}{1148} = 0.0000224$

Non-Urban:
$\frac{0.348^2 \times 0.0225 \times 0.9775}{1147} = 0.0000023$

Total: $Var = 0.0000247$

Precision Metrics

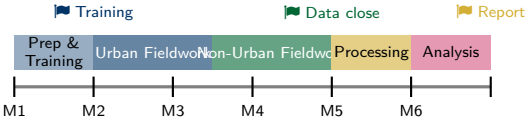
Standard Error:
$SE = \sqrt{0.0000247} = 0.00497$

Coefficient of Variation:
$CV = \frac{0.00497}{0.05} = 9.94\%$

95% Confidence Interval:
$0.05 \pm 1.96 \times 0.00497$
$= [4.03\%, 5.97\%]$

✔ CV target met!

Step 12: Implementation Timeline



Total duration: 6 months (within Ministry timeline)

Key milestones: M1 end—Interviewers trained; M4 end—All fieldwork complete; M6—Final report to Ministry

Step 13: Quality Assurance Plan

Field QA

- 10% back-checks on screened HH
- GPS verification of all visits
- Daily transmission of data
- Supervisor spot-checks
- Income verification protocol for borderline cases

Data QA

- Range checks on income
- Cross-validation with HH size
- Duplicate detection
- Weight trimming if needed
- Comparison with GHS benchmarks

Caution

Critical risk: Non-response among high-income HH may be higher than general population. Plan for 25% buffer in screening target (2,869 instead of 2,295).

Step 14: Risk Mitigation Strategies

Risk	Likelihood	Impact	Mitigation
High NR among wealthy HH	High	High	25% buffer; advance letters
Income underreporting	Medium	Medium	Indirect questions; asset proxies
Geographic clustering bias	Low	Medium	Sufficient PSU coverage
Screening errors	Medium	Low	Training; QC back-checks

Step 15: Final Budget Summary

Item	Quantity	Unit Cost	Total
Screening interviews	2,869	R50	R143,450
Full interviews	125	R500	R62,500
Training & preparation	Lump sum	—	R150,000
Supervision & QA	Lump sum	—	R200,000
Data processing	Lump sum	—	R100,000
Contingency (15%)	—	—	R98,393
TOTAL			**R754,343**

Budget: R1,500,000 | Cost: R754,343 | **Surplus: R745,657**

Step 16: Deliverables to Ministry

Survey Outputs Package

1. **Statistical Report**
   - Prevalence estimates with confidence intervals
   - Subgroup analysis by province and urban/rural
   - Income distribution among high-income households
   - Comparison with administrative tax data
2. **Microdata Package**
   - Anonymized household-level data
   - Sampling weights and replicate weights
   - Documentation and codebook
3. **Technical Documentation**
   - Sample design report
   - Weighting methodology
   - Quality indicators (response rates, CV by domain)

Case Study: Lessons Learned

1. Always profile the rare population BEFORE designing the sample
2. Geographic concentration enables efficient stratification
3. Screening dramatically reduces cost when rare trait is identifiable
4. Budget constraints often dictate design choices—quantify early
5. Plan for higher non-response among rare populations

Zambara Case

Lindiwe's reflection: "Lab 1.6 analysis was essential. Without it, I would have proposed an infeasible SRS design. Data-driven design saved the project—and R2.9 million."

Module 1.7 Complete: Rare Populations

Theory (241-

Why standard samplin
Sample size mathem
Screening approac
Oversampling appro
Decision framewo

Application (261-280)

Ministry brief analysis
Lab 1.6 results integration
Design specification
Weight calculation
Implementation plan

Lab 1.6: Hands-on calculations with GHS 2024 data

Day 1 Complete!
Tomorrow: Computational Tools and Spatial Integration

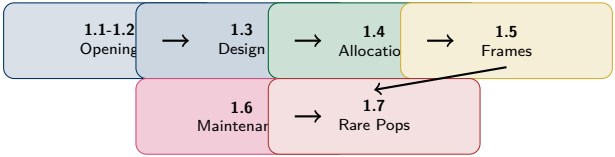
DAY 1 — CLOSING SESSION

Conclusion & The Day 1 Challenge

From Foundations to Application

Recap ★ Golden Rules Homework

Day 1: What We Covered



300 slides | 7 modules | 3 labs | 1 case study

Key Concepts: Design & Allocation

Multi-Stage Design

- Stratification ↓ variance
- Clustering ↑ variance
- $DEFF = 1 + (m-1)\rho$

GHS: DEFF=2.83, ρ=0.14

Allocation

- Proportional: $n_h \propto N_h$
- Neyman: $n_h \propto N_h S_h$
- Power: domain precision

Key: Match to objectives

Key Concepts: Frames & Rare Populations

Frame Issues

- Under/overcoverage
- Boundary changes
- GIS validation
- Listing updates

Zambara: 94 PSU problems fixed

Rare Populations

- SRS inefficient for P <10%
- Screening: 85% cost savings
- Oversampling: 68% reduction

Result: n=2,295 vs 7,299

The 10 Golden Rules

of Sample Design

★ ★ ★ ★ ★

Golden Rules 1–3

★ **Rule 1: Know Your Objective First**
Define target population, key indicators, required precision, and budget *before* designing.

★ **Rule 2: The Frame is the Foundation**
No sample can overcome a bad frame. Validate coverage before selection.

★ **Rule 3: Stratify to Reduce Variance**
Stratification never increases variance. If you can stratify, do it.

Golden Rules 4–6

★ **Rule 4: Clustering is a Necessary Evil**
Clusters save money but cost precision. Always calculate DEFF $= 1 + (m\text{-}1)\rho$.

★ **Rule 5: Allocation Should Match Objectives**
Proportional is rarely optimal. Use Neyman, power, or cost-optimal as needed.

★ **Rule 6: Every Weight Tells a Story**
Weights compensate for unequal selection. Unweighted estimates are BIASED.

Golden Rules 7–9

★ **Rule 7: Plan for What Can Go Wrong**
Hope for 100% response; plan for 70%. Build in buffers for non-response.

★ **Rule 8: Document Everything**
If it's not documented, it didn't happen. Your successor will thank you.

★ **Rule 9: Test Before You Deploy**
A R50K pilot can save a R5M disaster. Validate procedures in the field.

Golden Rule 10 + Summary

★ **Rule 10: Design for Analysis**
Provide users with PSU, stratum, and weight variables. A beautiful design with unusable data is failure.

Remember: Objective → Frame → Stratify → Cluster → Allocate → Weight → Document

Golden Rules: Quick Reference

THE 10 GOLDEN RULES

1. Know your objective
2. Frame is foundation
3. Stratify when possible
4. Clustering costs precision
5. Match allocation to goals
6. Weights tell stories
7. Plan for problems
8. Document everything
9. Test before deploy
10. Design for analysis

KEY FORMULAS

$$DEFF = 1 + (m-1)\rho$$

$$n_{eff} = n/DEFF$$

$$n_h = n \times \frac{N_h S_h}{\sum N_h S_h}$$

$$w = 1/\pi$$

HOMEWORK ASSIGNMENT

The Day 1 Challenge

Design a Sample for Your Country

Due: Tomorrow 08:30

The Day 1 Challenge

Assignment Brief

Scenario: Design a Labour Force Survey for your country.

Requirements:

- National unemployment estimate: CV $\leq$ 5%
- Provincial/regional estimates: CV $\leq$ 15%

Deliverable: 1-2 page design summary with calculations.

Key Insight

Tomorrow we compare designs across SADC member states!

Step 1: Gather Your Parameters

From your census or previous survey, find:

- ☑ Total households
- ☑ Unemployment rate (P)
- ☑ Typical cluster size (m)
- ☑ Number of provinces
- ☑ Urban/rural split
- ☑ Previous DEFF (if known)

If exact data unavailable, use reasonable estimates.

Step 2: Calculations Required

Task A: SRS Baseline

$$n = \frac{z^2 \times P \times (1-P)}{(CV \times P)^2}$$

Use $z = 1.96$, your country's unemployment rate P, CV=0.05

Task B: Design Effect Adjustment

$$DEFF = 1 + (m-1)\rho \qquad n_{adjusted} = n \times DEFF$$

Assume $m = 15$, $\rho = 0.05$ if unknown

Task C: Allocation Strategy
Propose allocation across provinces. Justify your choice.

Step 3: Submission Template

DAY 1 CHALLENGE

Country: _____ **Name:** _____

A. Parameters

HH: _____ P: _____ Provinces: _____

B. Calculations

n(SRS): _____ DEFF: _____ n(adj): _____

C. Allocation

Method: _____

Justification: _____

Submission Instructions

Deadline

Due: Day 2, 08:30
Format: 1-2 pages
Submit: To facilitators

Show your calculations!

Discussion

Tomorrow we will:

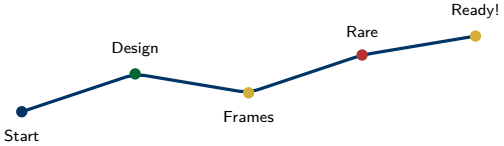
- Compare designs
- Share challenges
- Learn from peers

Note: No "wrong" answers—we want to see your reasoning!

Lindiwe's Day 1 Reflection

Zambara Case

Lindiwe: "What a journey! This morning I had a directive. Now I have a toolkit: design effects, optimal allocation, frame correction, and rare population strategies. Tomorrow, we code!"



Preview: Day 2 — Computational Tools

2.1-2.2 R Survey Package	**2.3** STATA svy
2.4 GIS Integration	**2.5** AI/ML Applications

BRING YOUR LAPTOP TOMORROW!

Thank You for Day 1!

See You Tomorrow

Day 2 begins at 08:30

Don't forget your homework!

Questions? See facilitators or email workshop@sadc-stats.org

SADC Regional Workshop Advanced Sampling Methods

Day 1 Complete

March 2026 | Johannesburg